



AMSC5720

Advanced Materials Synthesis

(先進材料合成)

Basic Concepts of Structure and Properties of Materials

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Topic Overview

- ◆ **Classification of Materials**
- ◆ **Building Blocks of Materials**
- ◆ **Structure of Crystalline Solids**
- ◆ **Imperfections in Solids**
- ◆ **Failure**

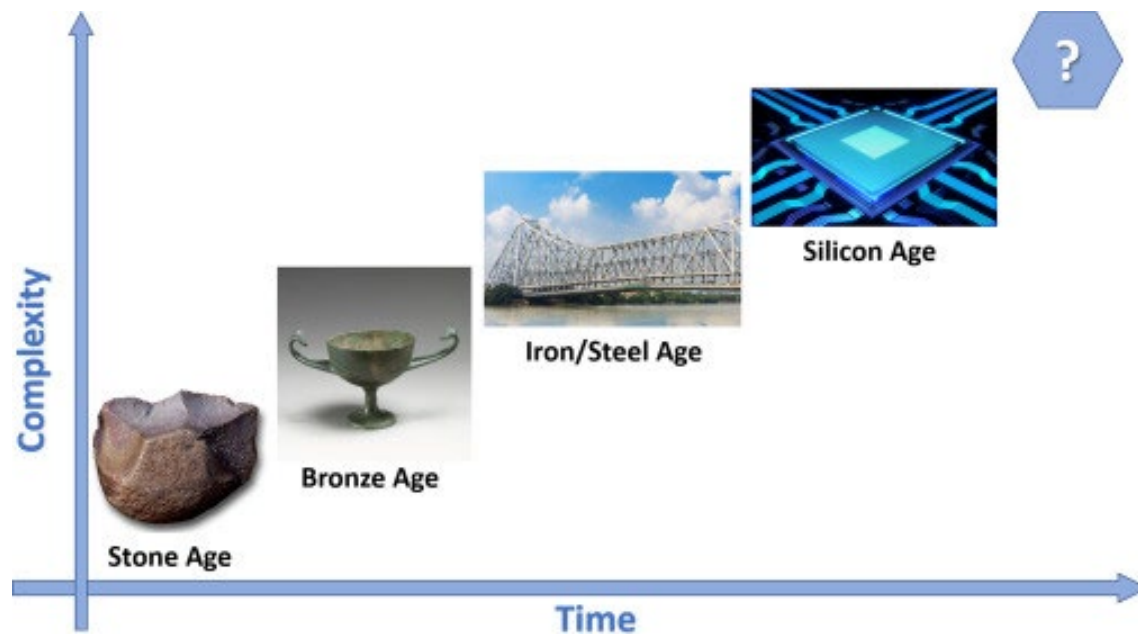
- Major textbook for this topic:

Materials Science and Engineering: An Introduction, by William D. Callister, David G. Rethwisch, 9th ed. 2014.



Why Materials Are Important

- Materials drive advancements in our society
 - Stone Age → Bronze Age → Iron Age



? → What is today's material age?

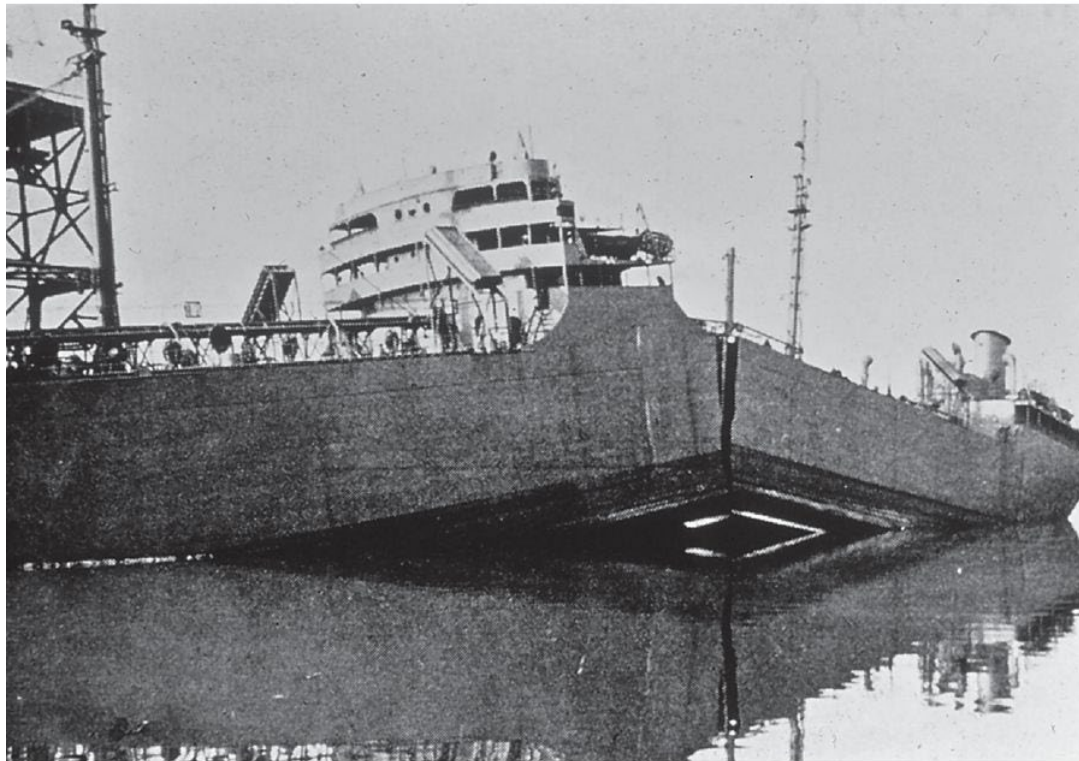
- Silicon Age (Electronic Materials)?
- Nanomaterials Age?
- Polymer Age?

<https://doi.org/10.1016/B978-0-12-818475-2.00019-2>



Why Materials Science Is Important

▣ The Liberty Ship Failure



Metal can be very ductile

<https://www.youtube.com/watch?v=bhL8WIDHKaY>

or very brittle

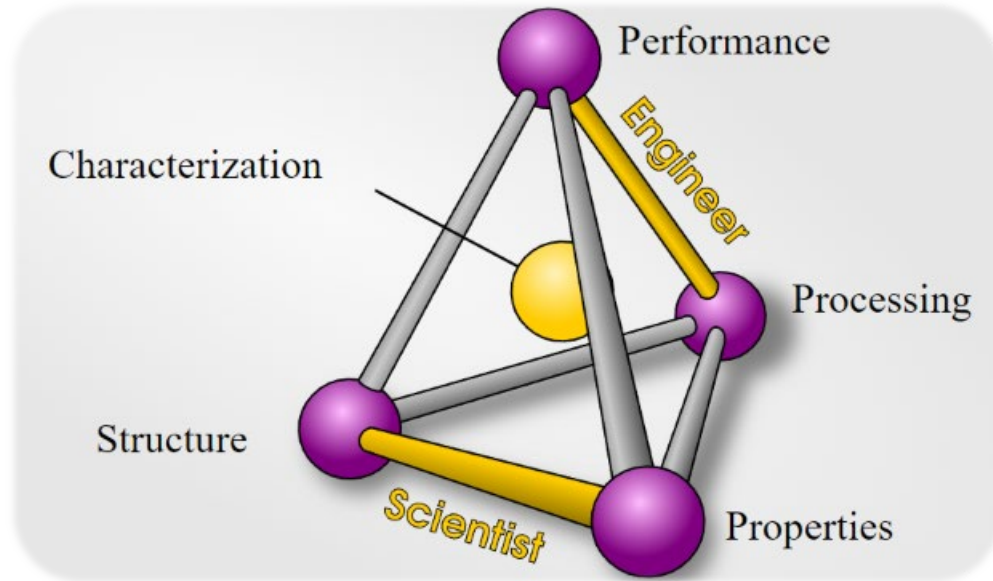
<https://www.youtube.com/watch?v=OSt11SA16KU>

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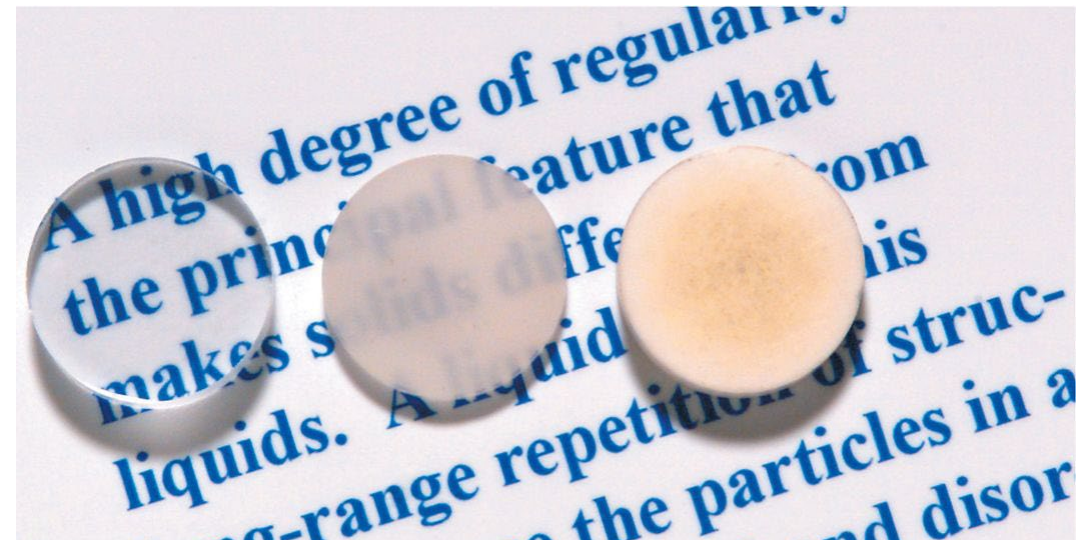


What is Materials Science

- ▣ **Materials science** investigates the *relationships* that *exist* between the *structures* and *properties* of materials.
- ▣ In other words: why materials have their properties.



"Materials tetrahedron"



William D. Callister Jr./ Specimen preparation, P.A. Lessing

Aluminum oxide with different structures show different light-transmittance characteristics.



What is Structure

- ▣ **Structure** can be a nebulous term
- ▣ It relates to the arrangement of a material's internal components
- ▣ Structural elements may be classified based on *size*:
 - » **Subatomic structure** (electrons within atoms, their energies and interactions with nuclei)
 - » **Atomic structure** (organization of atoms to form molecules/crystals)
 - » **Nanostructure** (aggregates of atoms to form nanoscale particles) < around 100 nm
 - » **Microstructure** (can be directly observed using optical microscopes) 100 nm to several mm
 - » **Macrostructure** (may be viewed by naked eyes) Several mm to m



What is Property

- ▣ **Property** describes the kind and magnitude of response of a material to a specific imposed stimulus
- ▣ All important properties of materials may be grouped into:
 - » **Mechanical properties** (deformations to an applied load)
 - » **Electrical properties** (stimulated by applied electric field)
 - » **Magnetic properties** (response to an applied magnetic field)
 - » **Thermal properties** (*e.g.* changes in temperature or temperature gradients)
 - » **Optical properties** (stimulated by electromagnetic or light radiation)
 - » **Deteriorative properties** (relate to chemical reactivity)
 - » **and so on...**

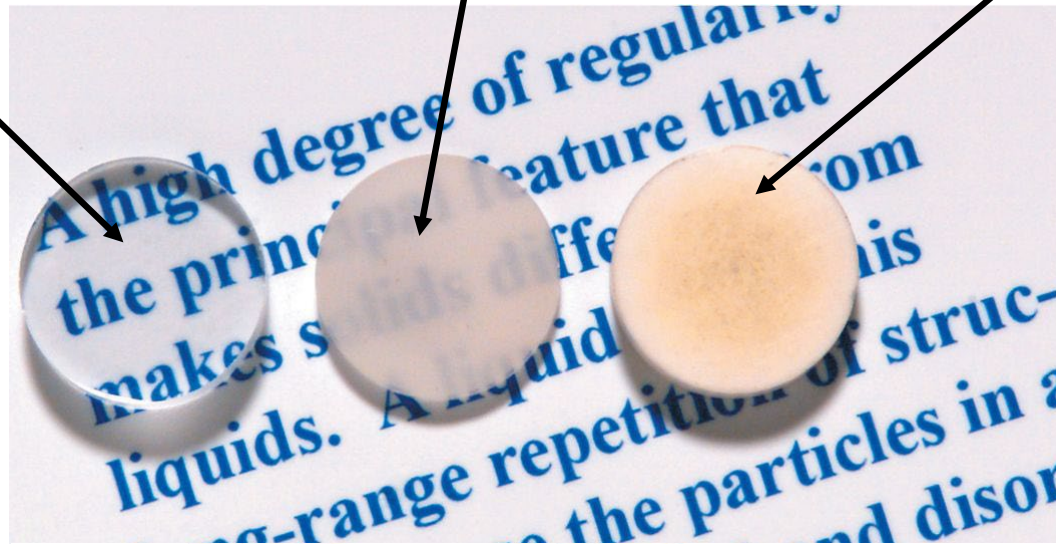


Example: optical property

Aluminum oxide **single crystal** (high degree of **perfection**)—is optically transparent

Aluminum oxide **polycrystalline material** (having many **small grains**)—is optically translucent

Aluminum oxide **polycrystalline material** having some **porosity**—is optically opaque



William D. Callister Jr./ Specimen preparation, P.A. Lessing



Classification of Materials

- ▣ **Metals** (no or small amounts of nonmetal elements)
 - Strong, ductile
 - High thermal & electrical conductivities
 - Opaque, reflective
- ▣ **Polymers/plastics** (often compounds of non-metal elements)
 - Soft, ductile, low strengths, low densities
 - Low thermal & electrical conductivities
 - Opaque, translucent or transparent
- ▣ **Ceramics** (often compounds of metal & non-metal elements)
 - Hard, Brittle
 - Low thermal & electrical conductivities
 - Opaque, translucent, or transparent



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© William D. Callister, Jr.

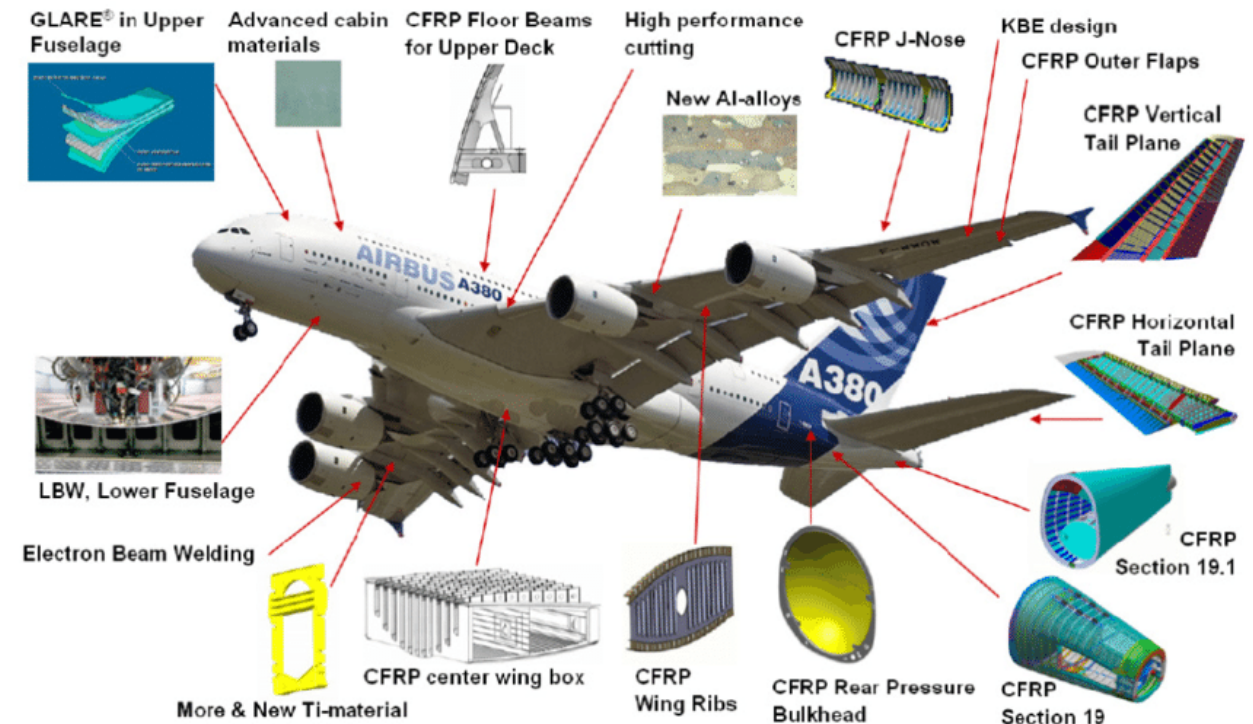
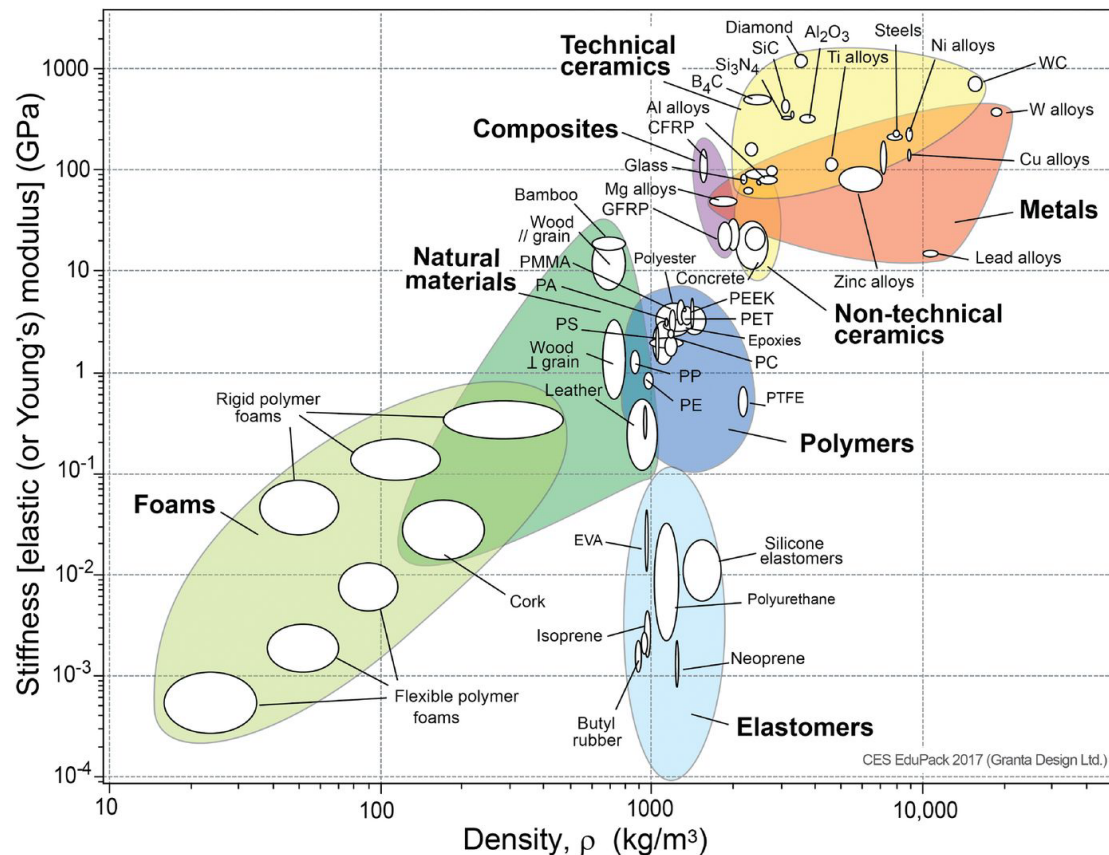


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Classification of Materials

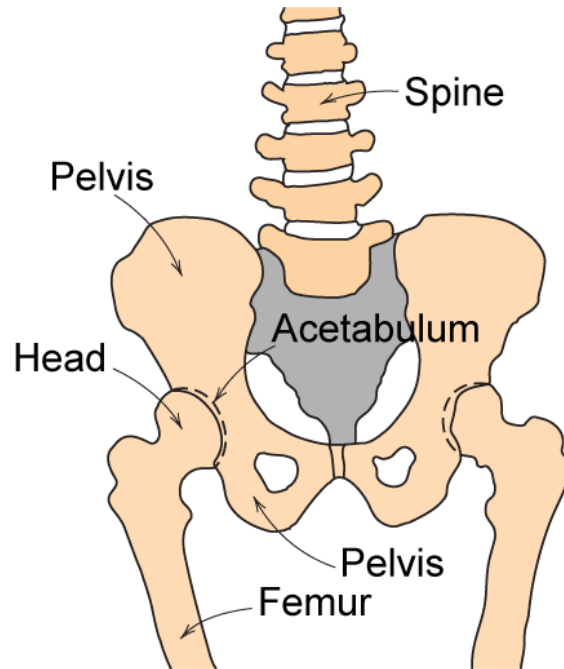
- **Composites** (more than one individual material)
 - Combination of properties that is not displayed by any single material



The AIRBUS A380 aircraft composite applications
[www.airbus.com]



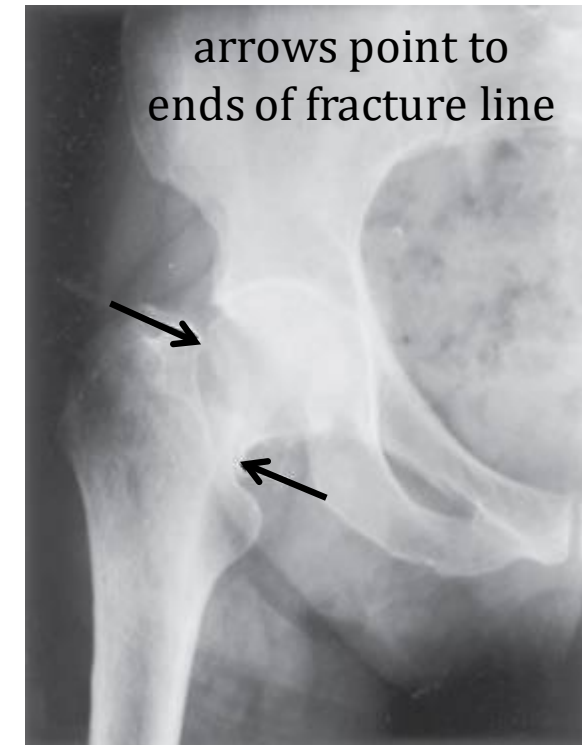
Example: Materials For Artificial Hip



Anatomy of a human hip joint and adjacent skeletal features



X-ray of normal hip joint



X-ray of fractured hip joint



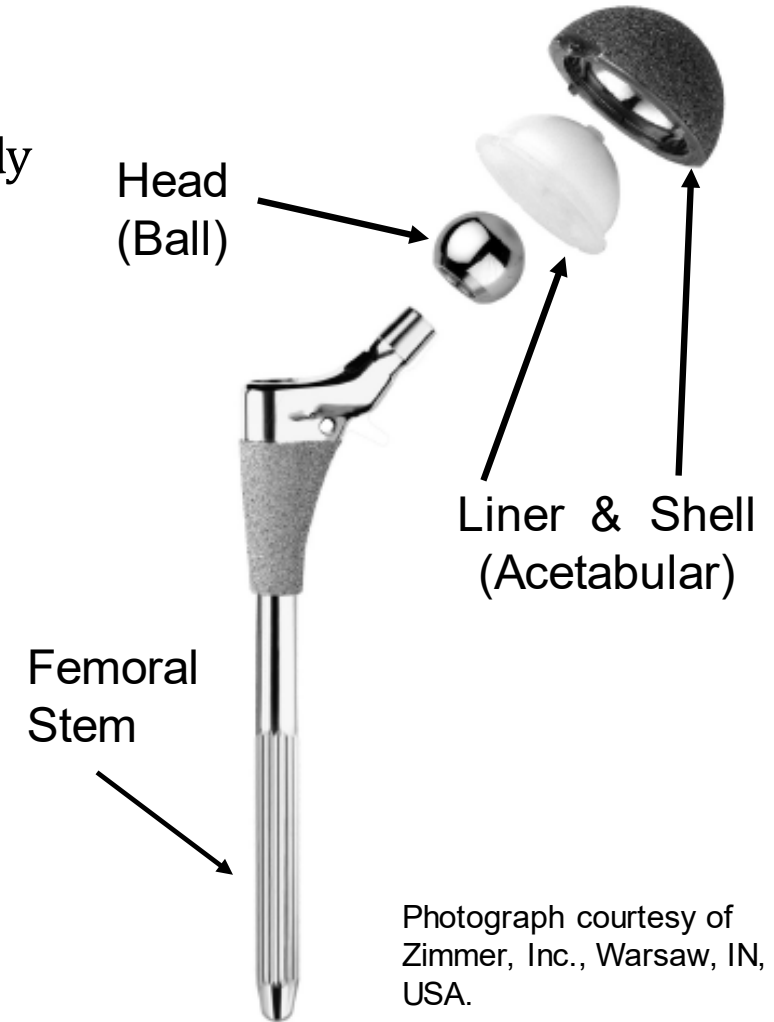
Example: Materials For Artificial Hip

▣ Materials requirements for artificial joints

- » **Biocompatible** – minimum rejection by surrounding body tissues
- » **Chemically inert** to body fluids
- » **Mechanical strength** to support forces generated
- » **Good lubricity** and **high wear resistance** between articulating surfaces

▣ Materials used

- » Femoral stem — titanium or CoCrMo alloy (**metal**)
- » Head (Ball) — CoCrMo alloy or Al_2O_3 (**ceramic**)
- » Shell — titanium alloy (**metal**)
- » Liner — polyethylene (**polymer**) or Al_2O_3 (**ceramic**)



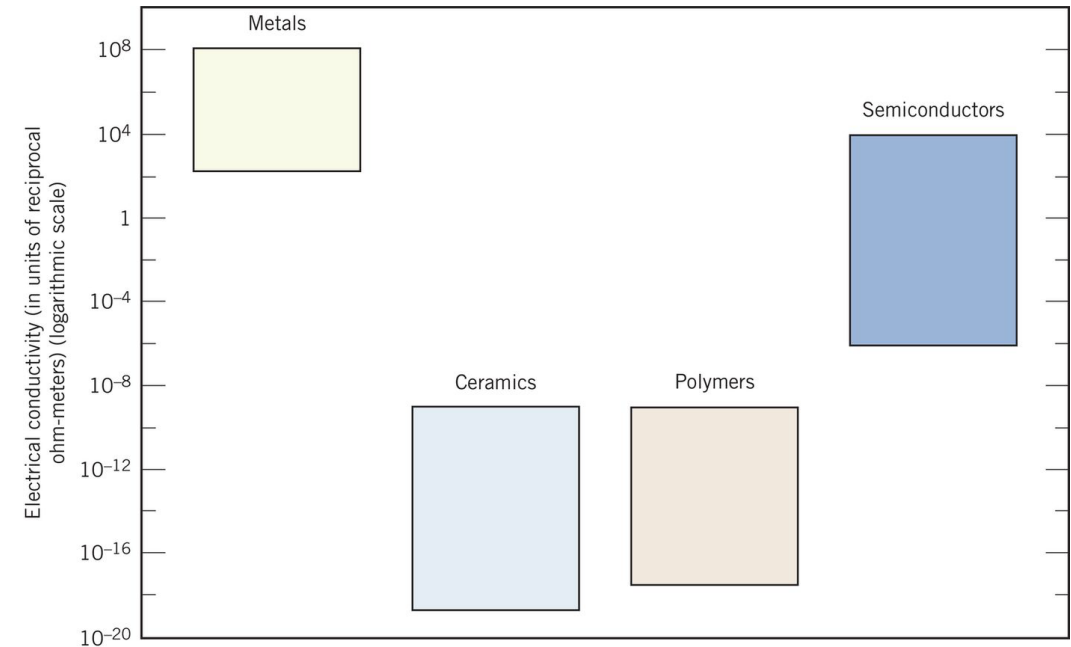
Photograph courtesy of Zimmer, Inc., Warsaw, IN, USA.



Advanced Materials

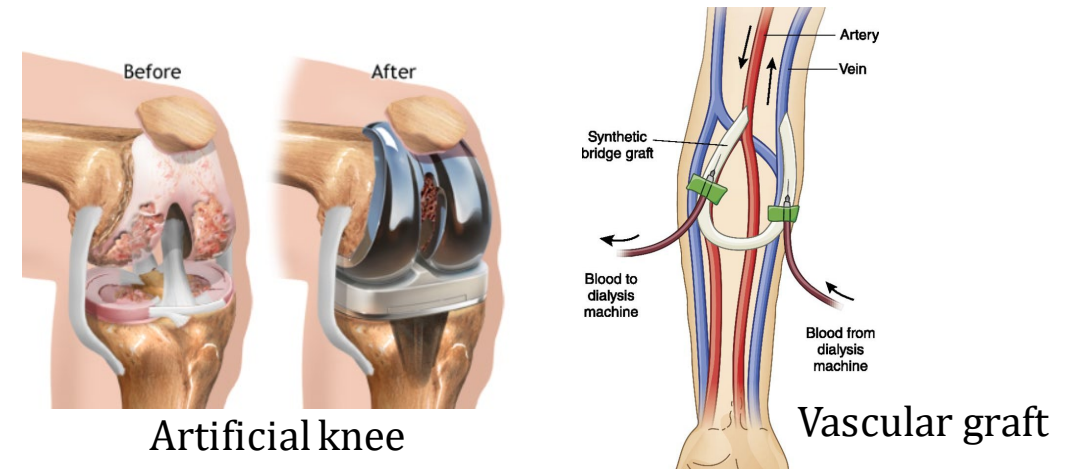
▣ Semiconductors

- Electrical properties are intermediate between conductors (metals and metal alloys) and insulators (ceramics and polymers)



▣ Biomaterials

- Biocompatible: compatible with body tissues and fluids
- Can be metals, ceramics, polymers, and composites





Advanced Materials

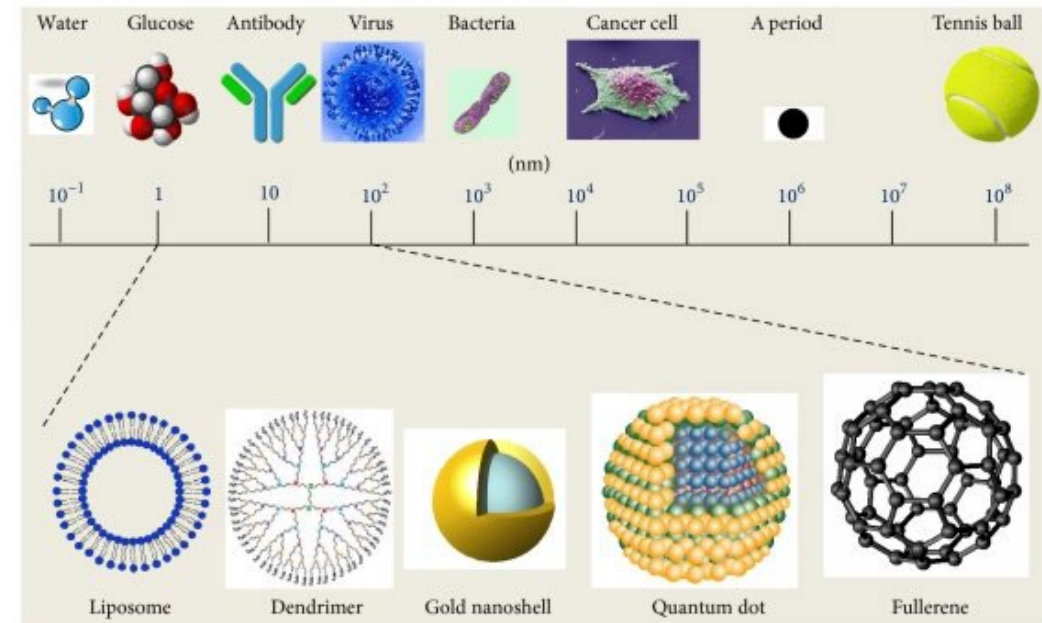
▣ Smart Materials

- Smart implies the ability to sense changes in environment and respond in predetermined manners
- e.g. shape-memory alloys, piezoelectric ceramics



▣ Nanomaterials

- Dimensions are on the order of 10^{-9} m
- Physical and chemical characteristics may experience dramatic changes
- Can be metals, ceramics, polymers, and composites





Examples: Different packaging materials for coke



© iStockphoto/Mark Oleksiy



© blickwinkel/Alamy



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Atomic Structure and Interatomic Bonding

- Understanding of interatomic interaction helps to explain a material's property:

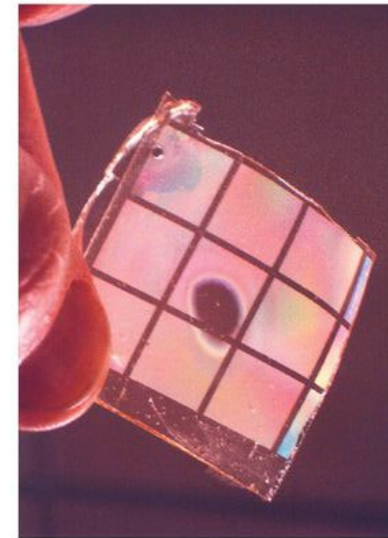
Barbara Peacock/Photodisc/Getty Images, Inc.



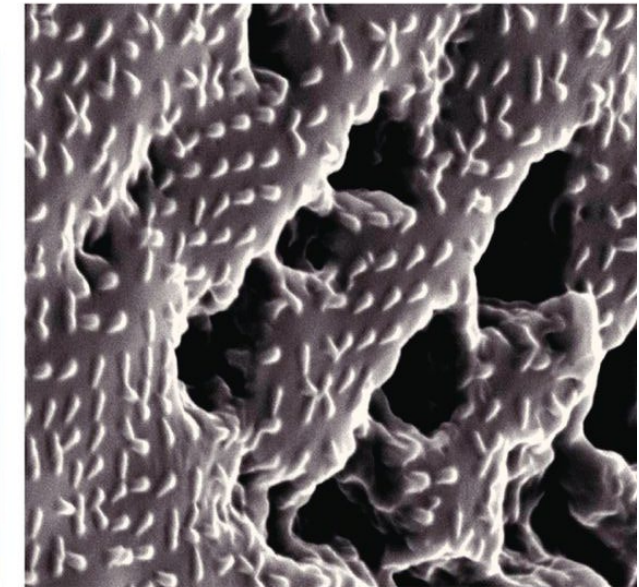
Paul D. Stewart/Science Source



Courtesy Jeffrey Karp,
Robert Langer and
Alex Galakatos



Courtesy Jeffrey Karp, Robert Langer
and Alex Galakatos



Gecko's very sticky feet $\xrightarrow{\text{Mechanism of adhesion}}$ *Synthetic adhesive*

<https://www.youtube.com/watch?v=YeSuQm7KfaE>



Atomic Structure

Let's recap some fundamental concepts

- ▣ Atom $\left\{ \begin{array}{l} \text{Electron: } 9.11 \times 10^{-31} \text{ kg} \\ \text{Proton: } 1.67 \times 10^{-27} \text{ kg} \\ \text{Neutron: } 1.67 \times 10^{-27} \text{ kg} \end{array} \right.$
- ▣ **Atomic number, Z** = no. of protons in nucleus = no. of electrons in neutral species
- ▣ **Atomic mass unit, amu** , is $1/12$ mass of ^{12}C
- ▣ **Atomic weight, A** , is weight of N_A (6.022×10^{23}) molecules or atoms
- ▣ $A \approx Z + N$, where N is no. of neutrons
- ▣ $1 \text{ amu/atom} = 1 \text{ g/mol}$



Electronic Structure

- ▣ Some of the chemical, electrical, thermal, & optical properties are determined by an atom's electronic structure
- ▣ Electrons have *wave-like* and *particle-like* characteristics
- ▣ Two wave-like characteristics are:
 - ▣ Electron position in terms of **probability density**
 - ▣ Shape, size, spatial orientation of probability density are determined by **quantum numbers**



Electronic Structure

TABLE 6.2 Relationship among Values of n , l , and m_l through $n = 4$

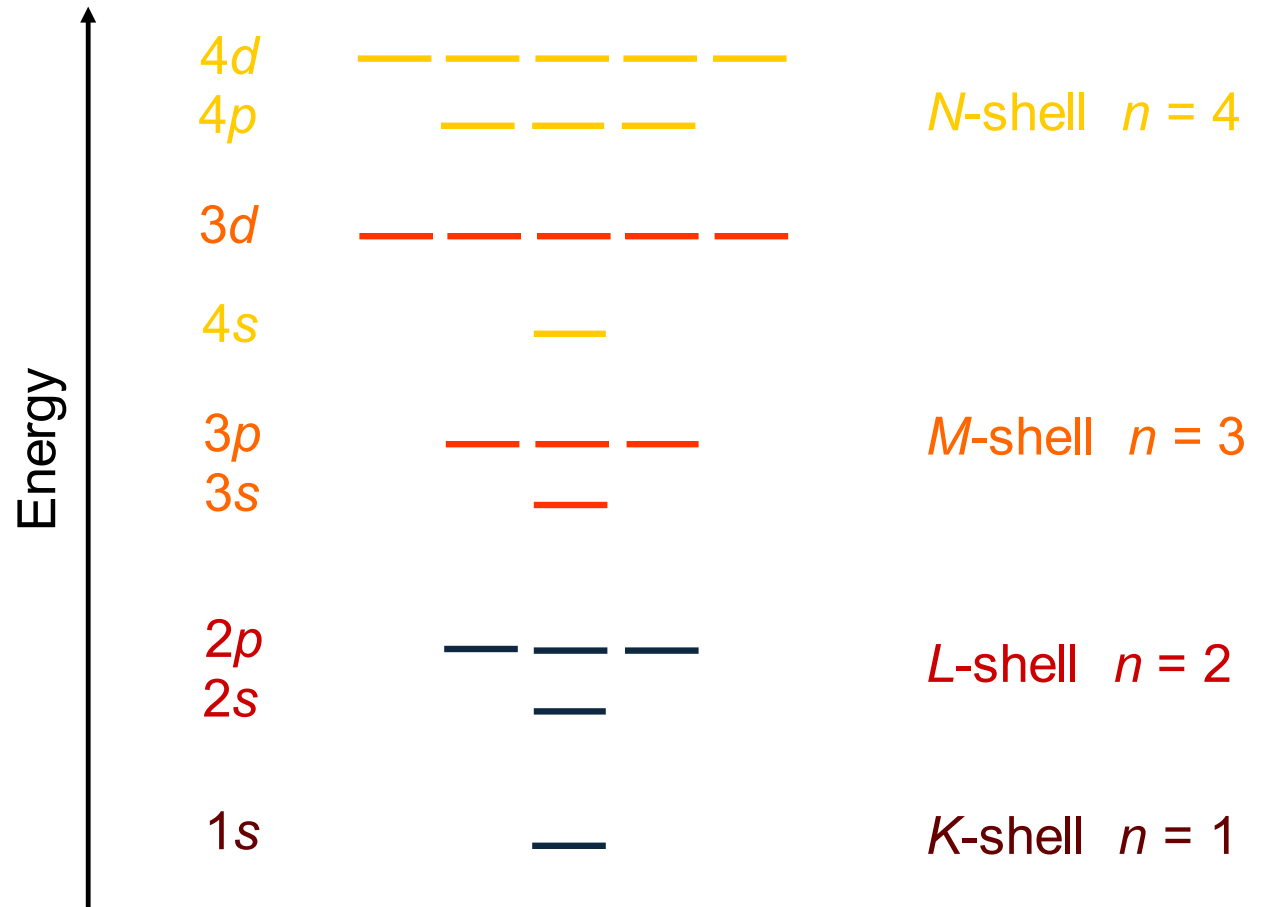
n	Possible Values of l	Subshell Designation	Possible Values of m_l	Number of Orbitals in Subshell	Total Number of Orbitals in Shell
1	0	1s	0	1	1
2	0	2s	0	1	4
	1	2p	1, 0, -1	3	
3	0	3s	0	1	9
	1	3p	1, 0, -1	3	
	2	3d	2, 1, 0, -1, -2	5	
4	0	4s	0	1	16
	1	4p	1, 0, -1	3	
	2	4d	2, 1, 0, -1, -2	5	
	3	4f	3, 2, 1, 0, -1, -2, -3	7	

(Chapter 6, Brown *et al.*)



Electron Energy States

- ▣ Electrons have **discrete** energy values
- ▣ Electrons tend to occupy **lowest available** energy states





Survey of Elements

- ▣ In most elements, electron configurations are **not stable**

Element	Atomic No.	Electron configuration
Hydrogen	1	$1s^1$
Helium	2	$1s^2$ (stable)
Lithium	3	$1s^2 2s^1$
Beryllium	4	$1s^2 2s^2$
Boron	5	$1s^2 2s^2 2p^1$
Carbon	6	$1s^2 2s^2 2p^2$
...	...	...
Neon	10	$1s^2 2s^2 2p^6$ (stable)
Sodium	11	$1s^2 2s^2 2p^6 3s^1$
Magnesium	12	$1s^2 2s^2 2p^6 3s^2$
Aluminum	13	$1s^2 2s^2 2p^6 3s^2 3p^1$
...	...	...
Argon	18	$1s^2 2s^2 2p^6 3s^2 3p^6$ (stable)
...	...	...
Krypton	36	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$ (stable)

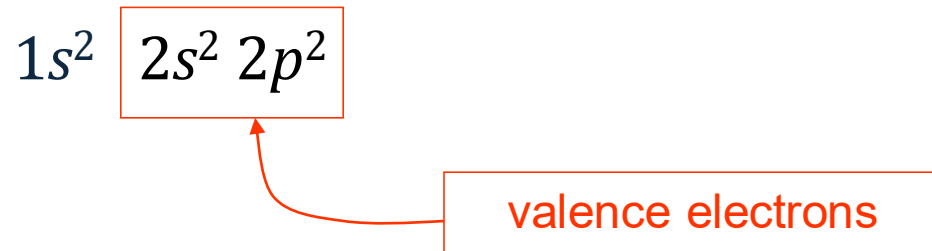
Why not stable?

➤ **Valence** (outer) shell usually **not** completely **filled**.



Electronic Configurations

- ▣ **Valence electrons** – electrons in outer unfilled shells
- ▣ Filled shells are more stable – require more energy to gain or lose electrons
- ▣ Valence electrons are **available for bonding** and tend to determine an atom's chemical properties
 - ▣ example: C (atomic number = 6)





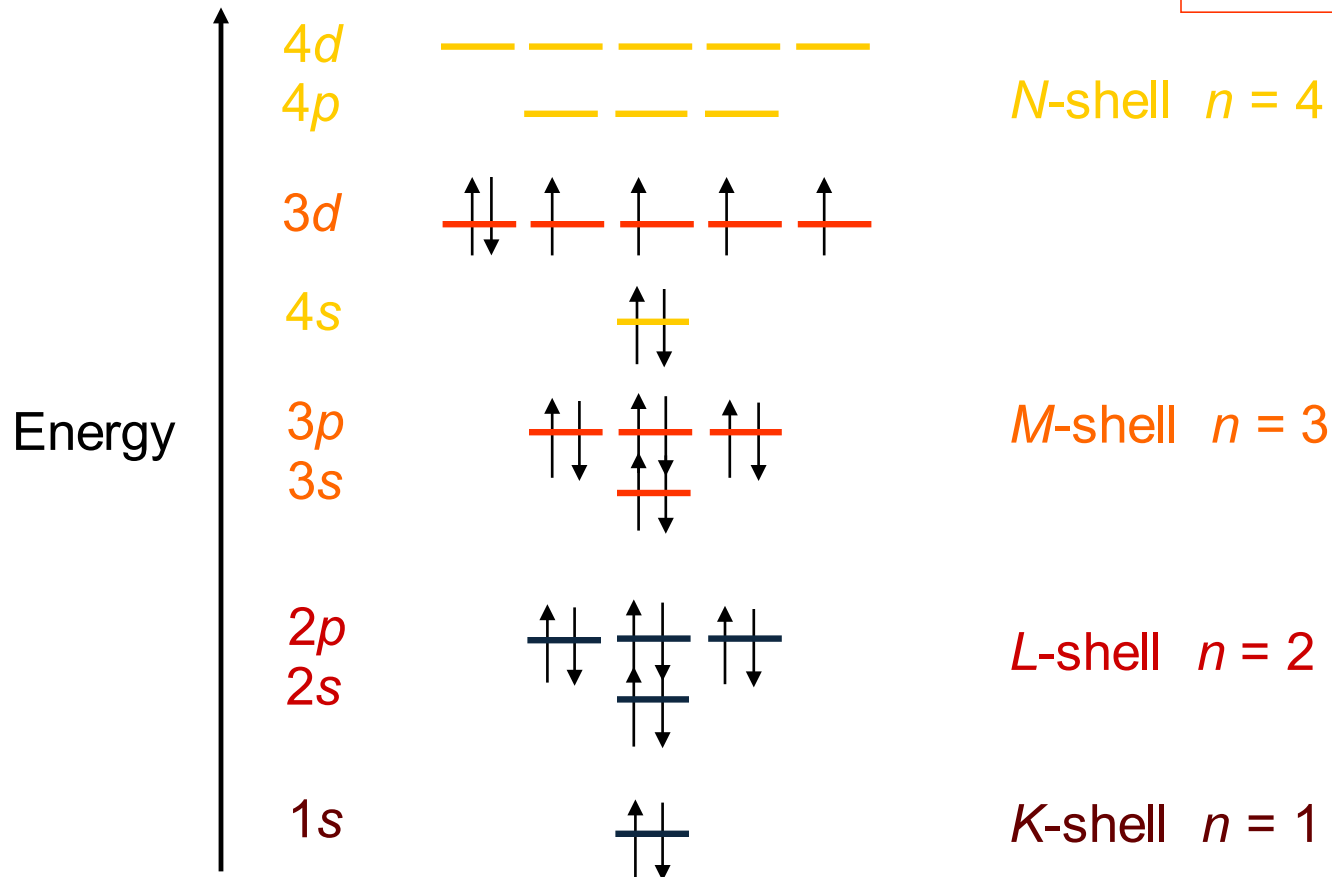
Electronic Configurations

- Example: Fe (atomic number = 26)

Electron configuration

$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$

valence
electrons

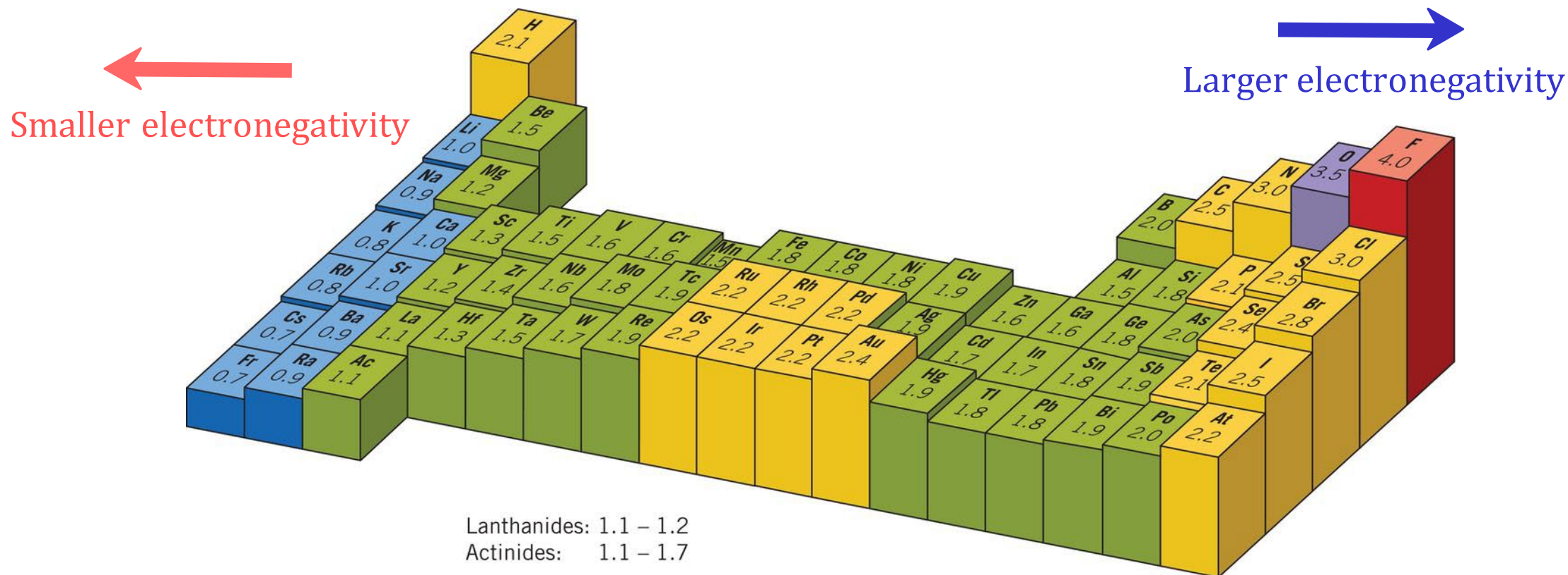


- ✓ Transition metals can use the electrons in their **d** subshells as valence electrons.



Electronegativity

- ▣ Ability of atoms to *attract* electrons, ranges from 0.7 to 4.0
- ▣ Larger electronegativity → higher tendency to acquire electrons

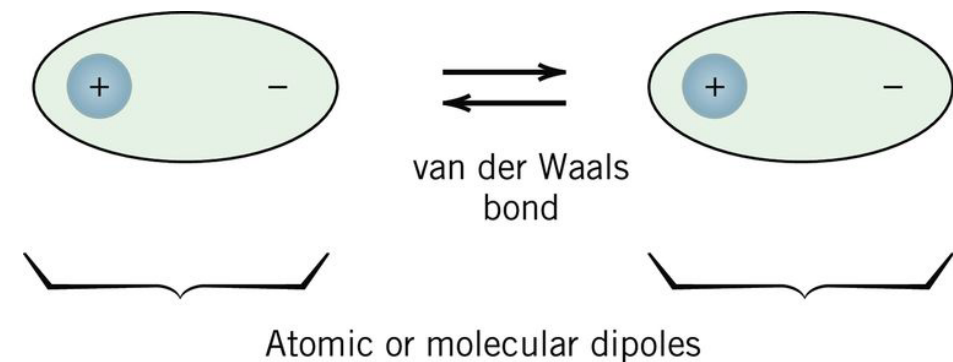


From J. E. Brady and F. Senese, *Chemistry: Matter and Its Changes*, 4th edition, 2004. This material is reproduced with permission of John Wiley & Sons, Inc.



Atomic Bonding in Solids

- ▣ **Primary** Interatomic Bonds
 - ▣ Ionic bonding
 - ▣ Covalent bonding
 - ▣ Metallic bonding
- ▣ **Secondary** Bonding or **van der Waals** Bonding (arisen from atomic or molecular dipoles)
 - ▣ Permanent dipole bonds
 - ▣ Hydrogen bonding
 - ▣ Fluctuating induced dipole bonds
 - ▣ Polar molecule-induced dipole bonds





Ionization Process

▣ Main text

metal atom + nonmetal atom



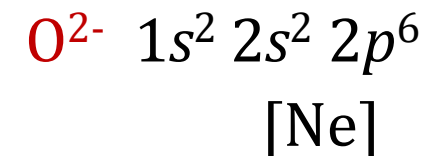
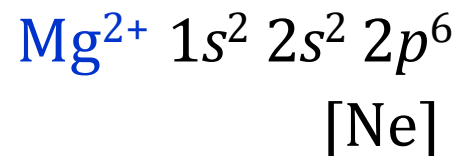
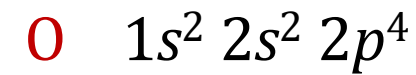
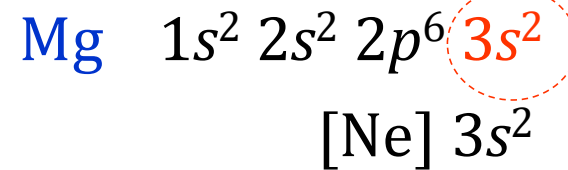
donates
electrons



accepts
electrons

Dissimilar electronegativities:

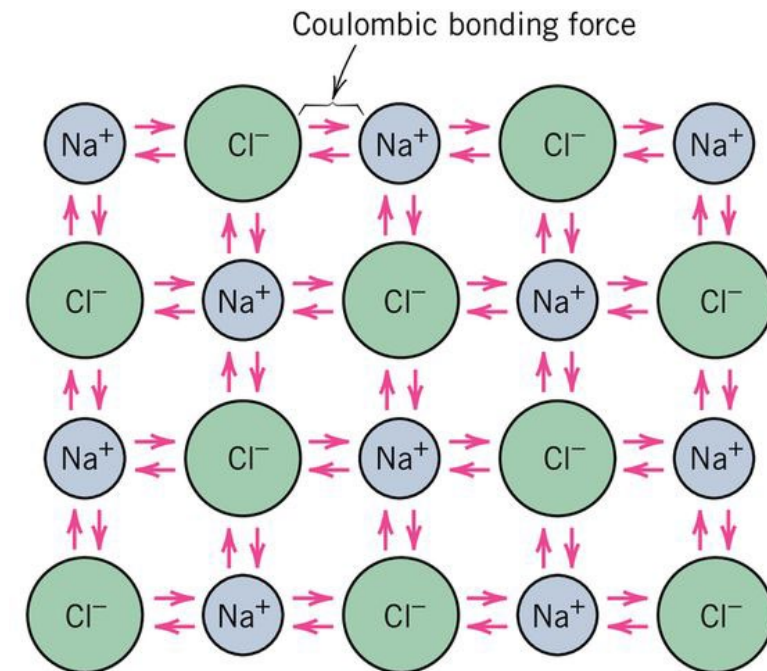
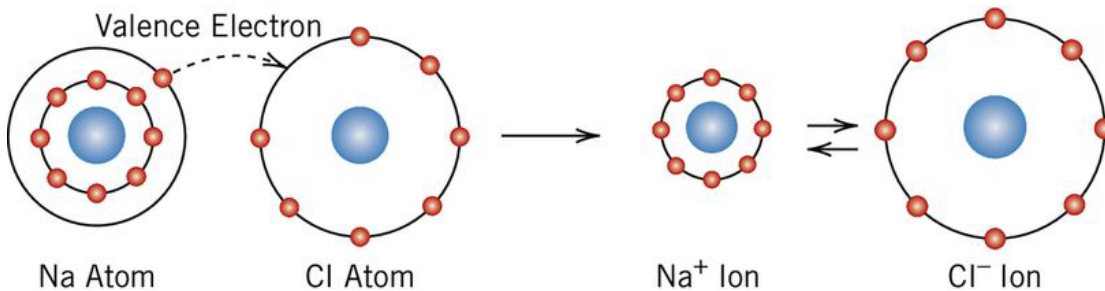
example: MgO





Ionic Bonding

- Occurs between positive and negative ions (often metal + nonmetal)
- Requires **electron transfer** → often achieves **inert gas configurations**
- Requires **large** difference in electronegativity
 - Example: NaCl

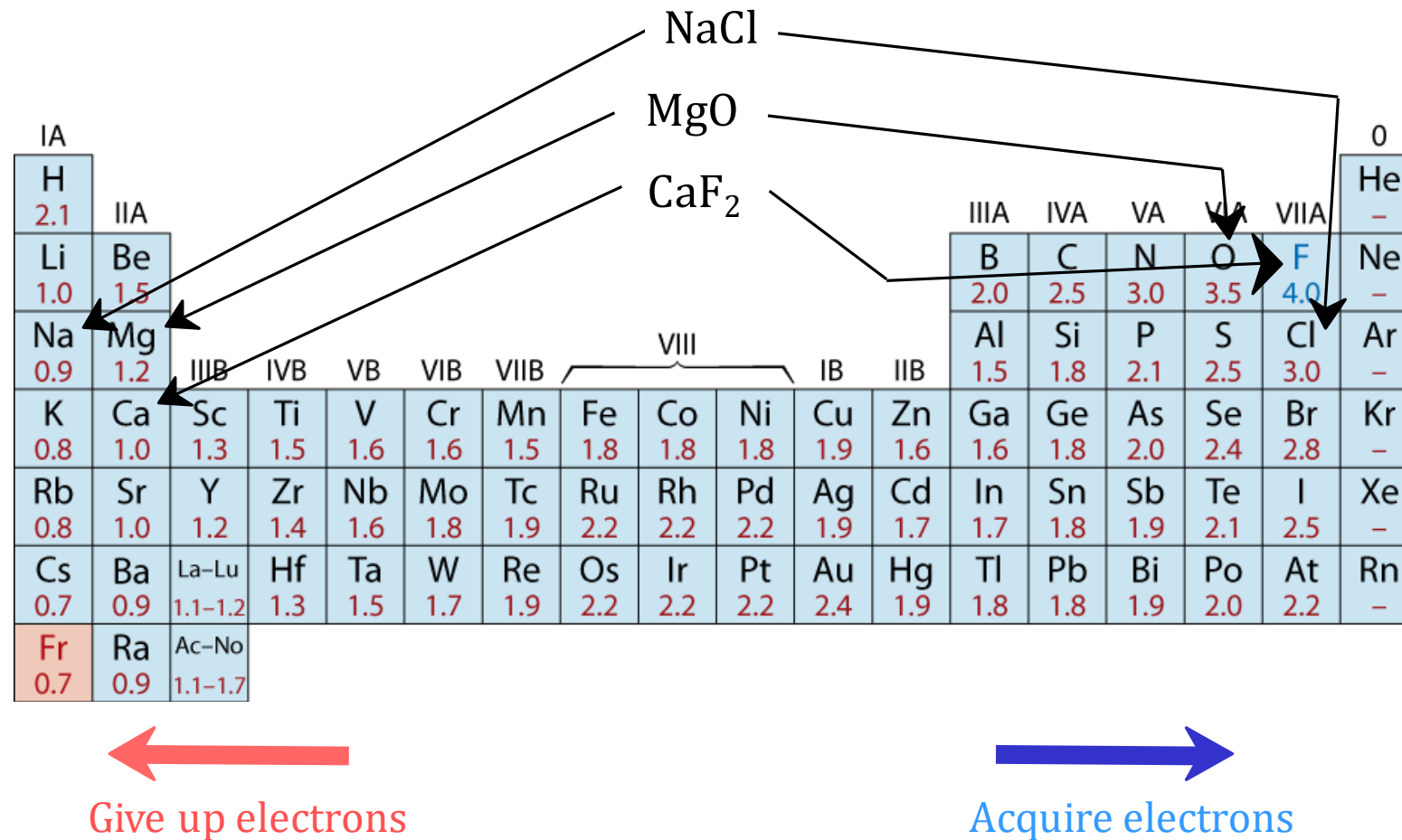




Ionic Bonding

▣ Predominant bonding in **Ceramics**

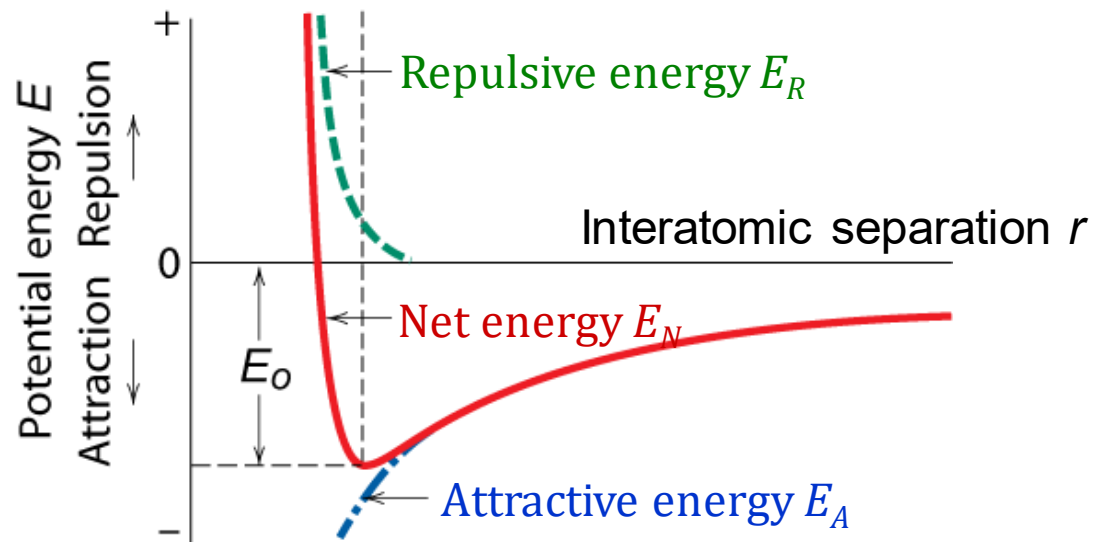
Examples:





Ionic Bonding

- ▣ Net energy (E_N) = sum of attractive and repulsive energies
- ▣ Equilibrium separation when net energy is at minimum (E_0)
- ▣ **Bonding energy:** minimum energy to separate two atoms to an infinite distance

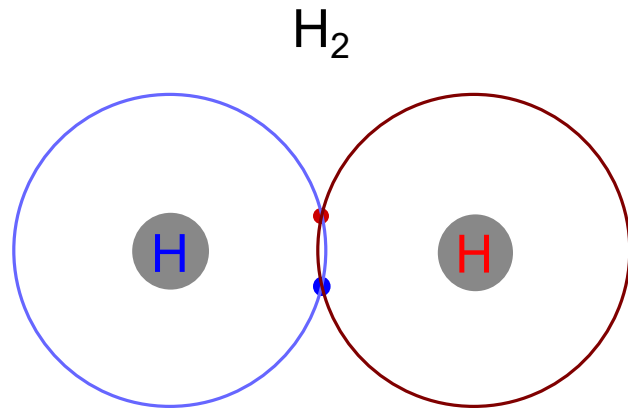


$$E_N = E_A + E_R$$



Covalent Bonding

- ▣ Atoms have **small** differences in electronegativity
- ▣ Stable electron configurations are assumed by **sharing of valence e^-**
- ▣ Normally s and p orbitals are involved
 - ▣ Example:



- shared 1s electron from 1st hydrogen atom

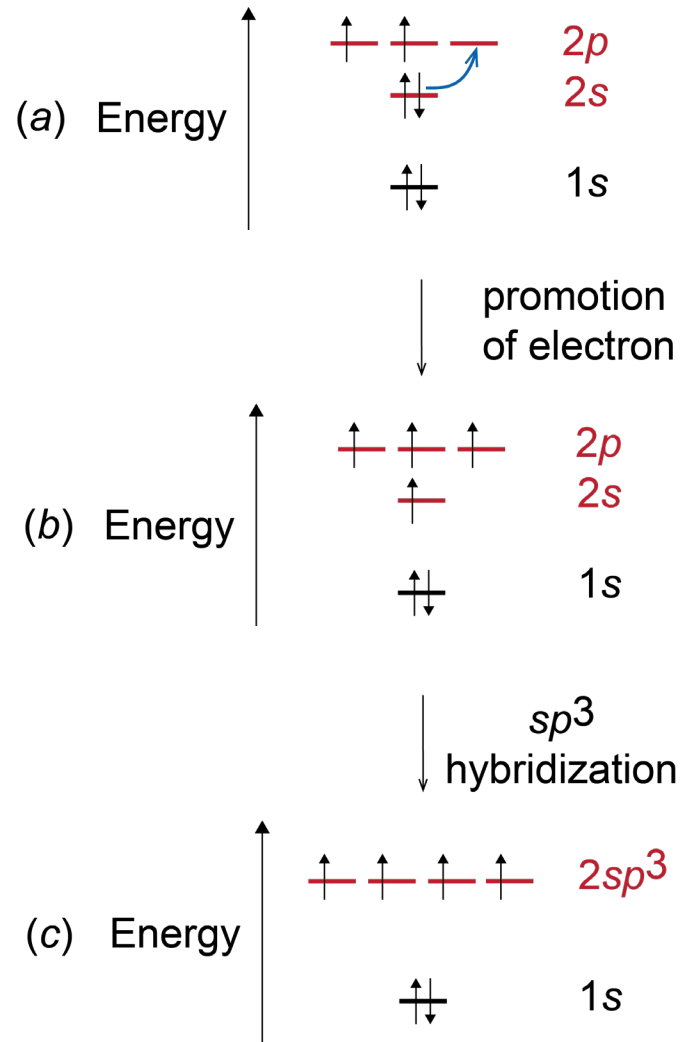
- shared 1s electron from 2nd hydrogen atom

- ✓ Each H has 1 valence e^- , needs 1 more
- ✓ Electronegativities are the same

Fig. 2.12, Calliser & Rethwisch 10e.



Bond Hybridization in Covalent Bonding



- Example : carbon can form sp^3 hybrid orbitals

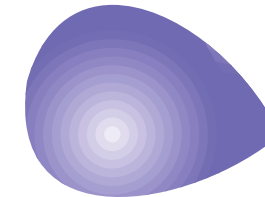


Fig. 2.14, Callister & Rethwisch 10e.

(Adapted from J.E. Brady and F. Senese, *Chemistry: Matter and Its Changes*, 4th edition. Reprinted with permission of John Wiley and Sons, Inc.)



Bond Hybridization in Covalent Bonding

Hybrid sp^3 bonding involving carbon

Example: CH_4

- ✓ Each C has 4 valence e^- , needs 4 more
- ✓ Each H has 1 valence e^- , needs 1 more

Electrons are shared in sp^3 hybrid covalent bonds.

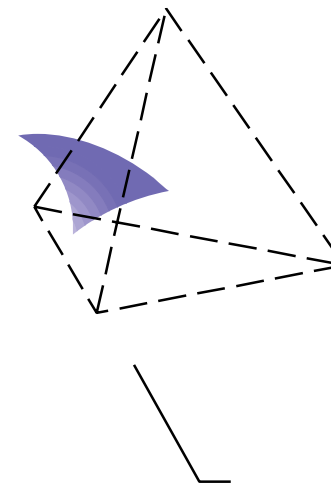


Fig. 2.15, Callister & Rethwisch 10e.
(Adapted from J.E. Brady and F. Senese, *Chemistry: Matter and Its Changes*,
4th edition. Reprinted with permission of John Wiley and Sons, Inc.)



Metallic Bonding

- ▣ Seen in metals and their alloys
- ▣ Electrons delocalized to form an “**electron cloud**” or “**sea of electrons**”
- ▣ Free electrons act as “glue” to hold ion cores together

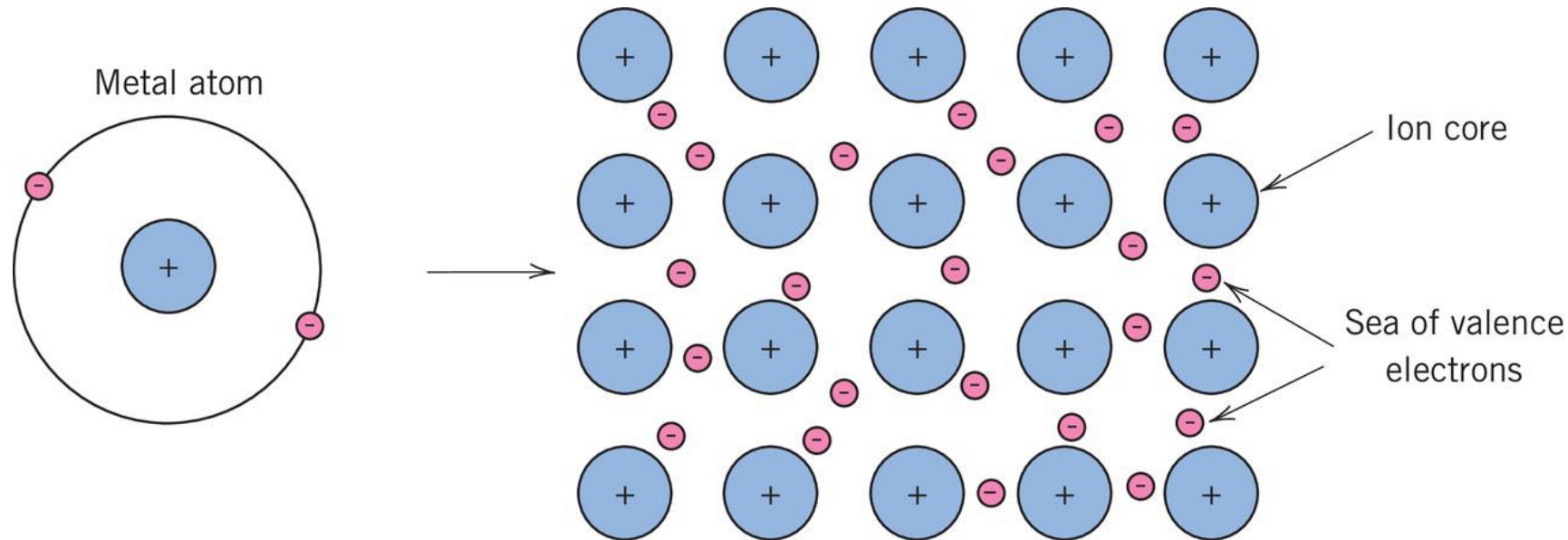


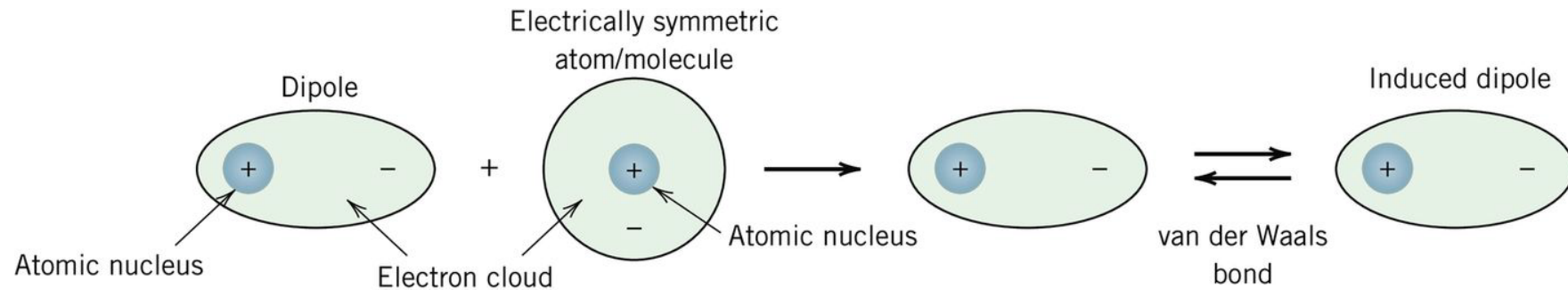
Fig. 2.19, *Callister & Rethwisch 10e.*



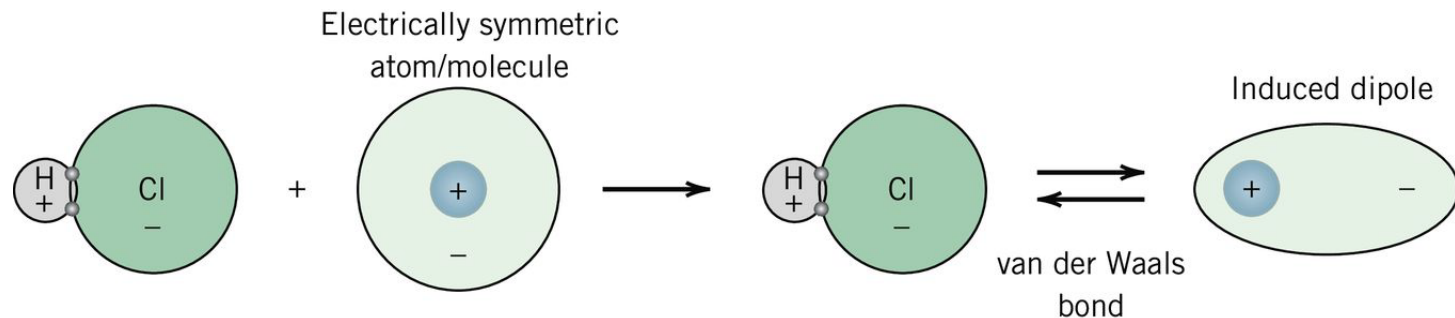
Secondary Bonding

- Arises from attractive forces between dipoles

- Fluctuating dipoles

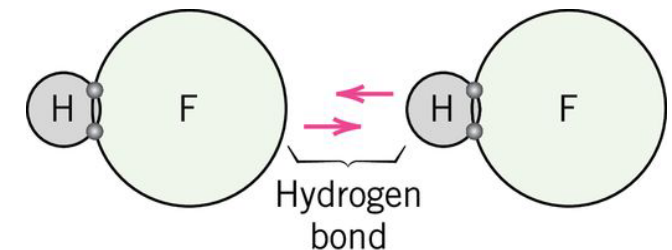


- Polar molecule-induced dipole bonds



- Permanent dipoles

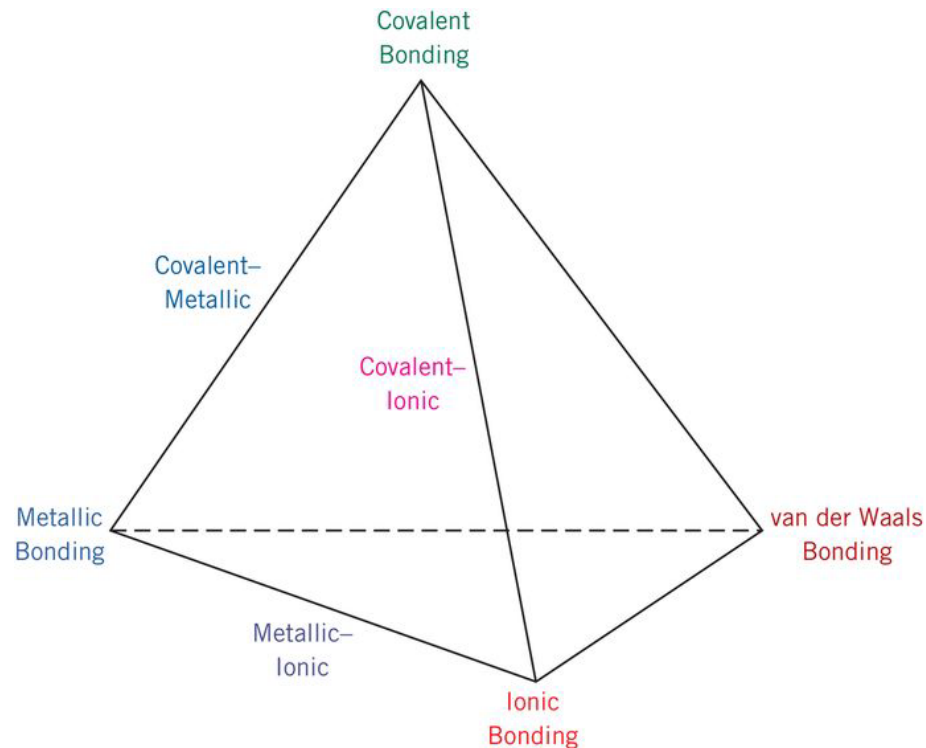
- Example: **Hydrogen bond**



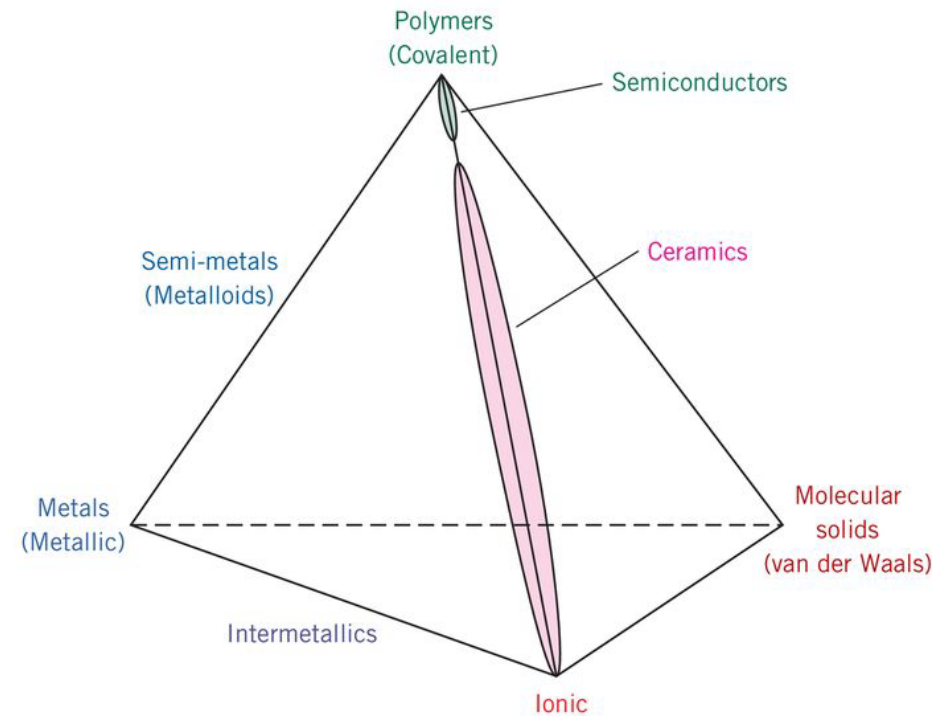


Mixed Bonding

- In real materials, the bonds are sometimes *mixtures* of several bond types
- We can use a bonding tetrahedron to illustrate



“Bonding tetrahedron”



“Material-type tetrahedron”



Mixed Bonding

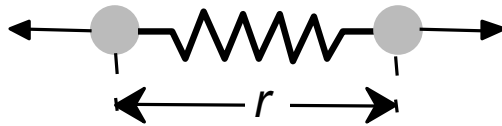
- ▣ Most common mixed bonding is **mixed covalent-ionic bonds**
- ▣ **Percent ionic character (%IC)** of a bond between elements A and B (A being the most electronegative) may be approximated by:
 - ▣ X_A and X_B being the electronegativities of the respective elements
 - ▣ *Example:* compute the percent ionic character (IC%) of the interatomic bond that forms between C and H, given X_C is 2.5 and X_H is 2.1:
- ▣ Thus the C–H bond is primarily covalent (96.1%)



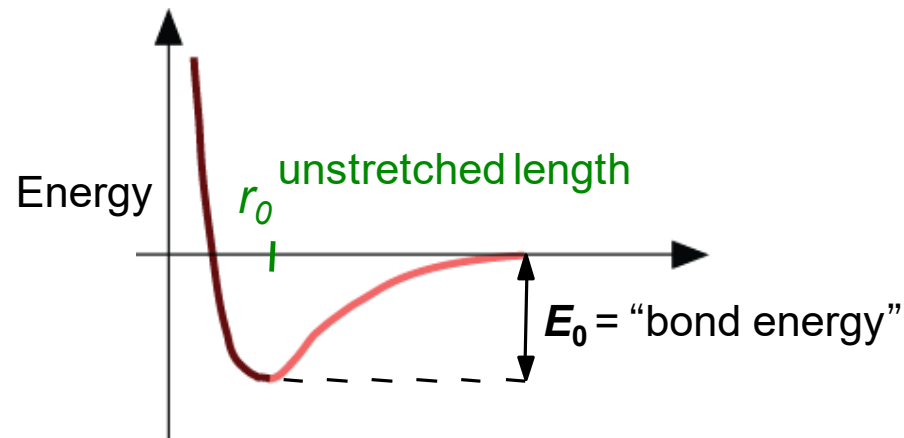
Properties Related to Bonding

- Many physical and chemical properties of materials are closely related to the information from *E - r curve*

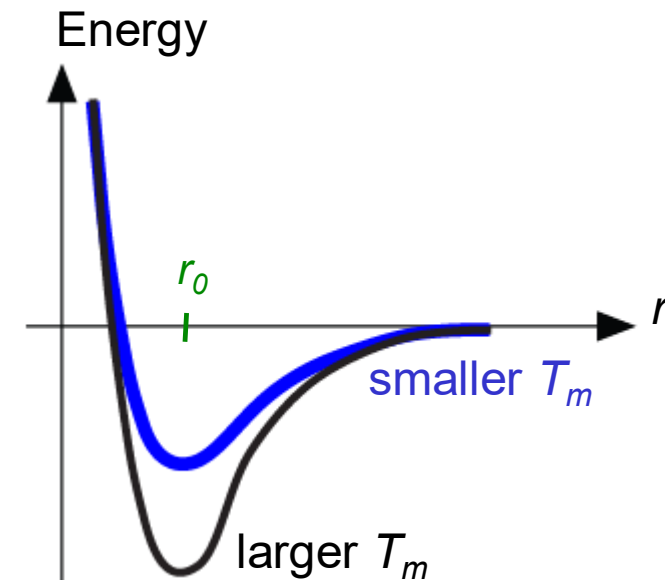
- Bond length, r



- Bond energy, E_0



- Melting temperature (T_m):
higher $E_0 \rightarrow$ higher T_m





Bonding Energy & T_m

Table 2.3

Bonding Energies and
Melting Temperatures
for Various Substances

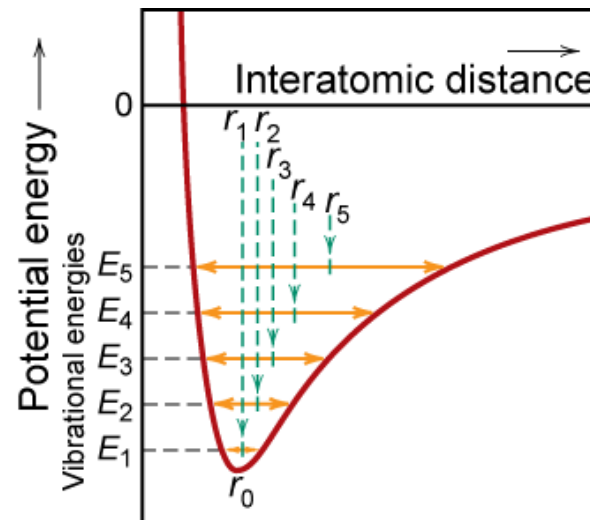
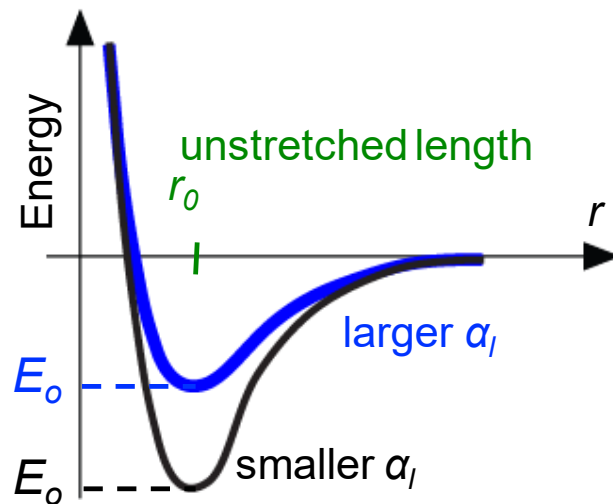
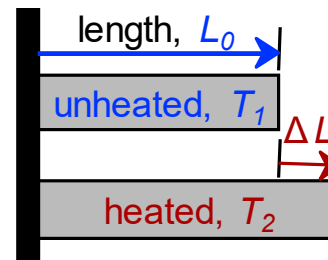
<i>Substance</i>	<i>Bonding Energy (kJ/mol)</i>	<i>Melting Temperature (°C)</i>
Ionic		
NaCl	640	801
LiF	850	848
MgO	1000	2800
CaF ₂	1548	1418
Covalent		
Cl ₂	121	−102
Si	450	1410
InSb	523	942
C (diamond)	713	>3550
SiC	1230	2830
Metallic		
Hg	62	−39
Al	330	660
Ag	285	962
W	850	3414
van der Waals^a		
Ar	7.7	−189 (@ 69 kPa)
Kr	11.7	−158 (@ 73.2 kPa)
CH ₄	18	−182
Cl ₂	31	−101



Properties Related to Bonding

- Linear coefficient of thermal expansion (α_l): deeper/narrower shape of the curve (typically means larger E_0) $\rightarrow$ lower α_l (smaller dimensional alterations upon T change)

$$\frac{\Delta L}{L_0} = \alpha_l (T_2 - T_1)$$



- Increase in bond length is due to **asymmetry** of the ***E-r curve***. This results in an increase in α_l .
- As E_0 increases this asymmetry decreases.



Properties Related to Bonding

Ceramics

Ionic & covalent bonding

Large bond energy

- high T_m
- small α_l

Metals

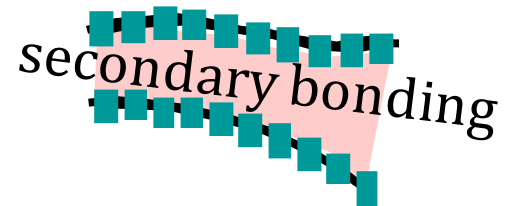
Metallic bonding

Variable bond energy

- moderate T_m
- moderate α_l

Polymers

Covalent & Secondary



Weak bond energy (between chains)
Secondary bonding responsible for most physical properties

- low T_m
- large α_l



Brief Summary

- ▣ A material's chemical, electrical, thermal, and optical properties are determined by **electronic configuration**.
- ▣ **Valence electrons** occupy the outermost unfilled electron shell.
- ▣ **Primary bonding** types include covalent, ionic, and metallic bonding.
- ▣ **Secondary or van der Waals bonds** are weaker than the primary bonding types.
- ▣ In real materials, bonding type can be **mixed**, an estimated **percent ionic character** may be calculated
- ▣ Some physicochemical properties can be deduced from $E-r$ curve



The Structure of Crystalline Solids

Let's step into the next level of the structure of materials: **arrangement of atoms** in solid state

- ▣ Fundamental concepts to cover in this topic:
 - Crystallinity vs. noncrystallinity
 - Crystal structure
 - Crystallographic points
 - Directions
 - Planes
 - Single crystal vs. polycrystalline materials
 - Some important crystal structures



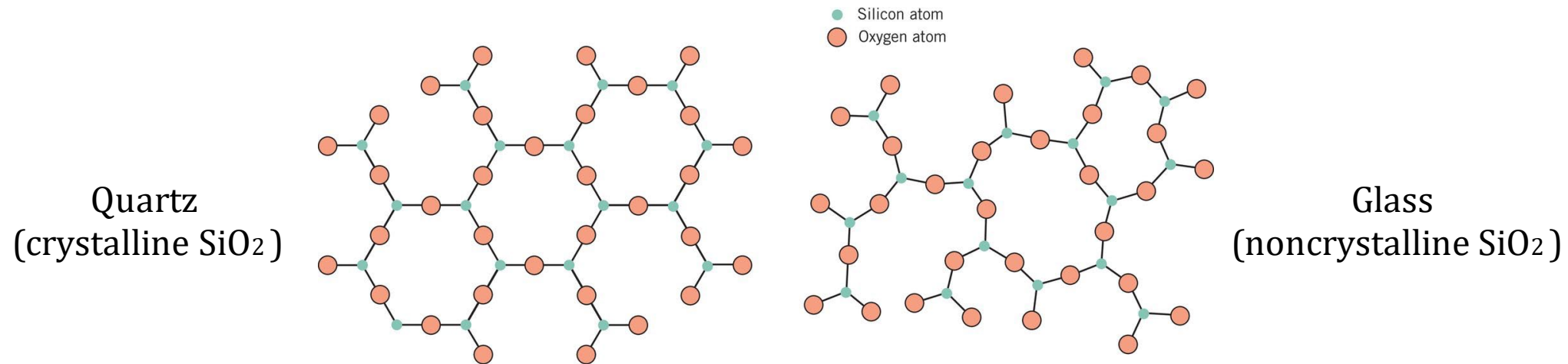
Crystalline vs. Noncrystalline

- ▣ A **crystalline** material is one in which the atoms are situated in a *repeating* or *periodic* array over large atomic distances
- ▣ That is, **long-range** order exists, such that upon solidification, atoms will position themselves in a **repetitive three-dimensional** pattern
- ▣ All metals, many ceramics, and certain polymers are crystalline under normal solidification conditions
- ▣ A **noncrystalline** (or **amorphous**) material do not crystalline, that is, long-range atomic order is **absent**



Crystalline vs. Noncrystalline

- Some substances have both crystalline or noncrystalline forms

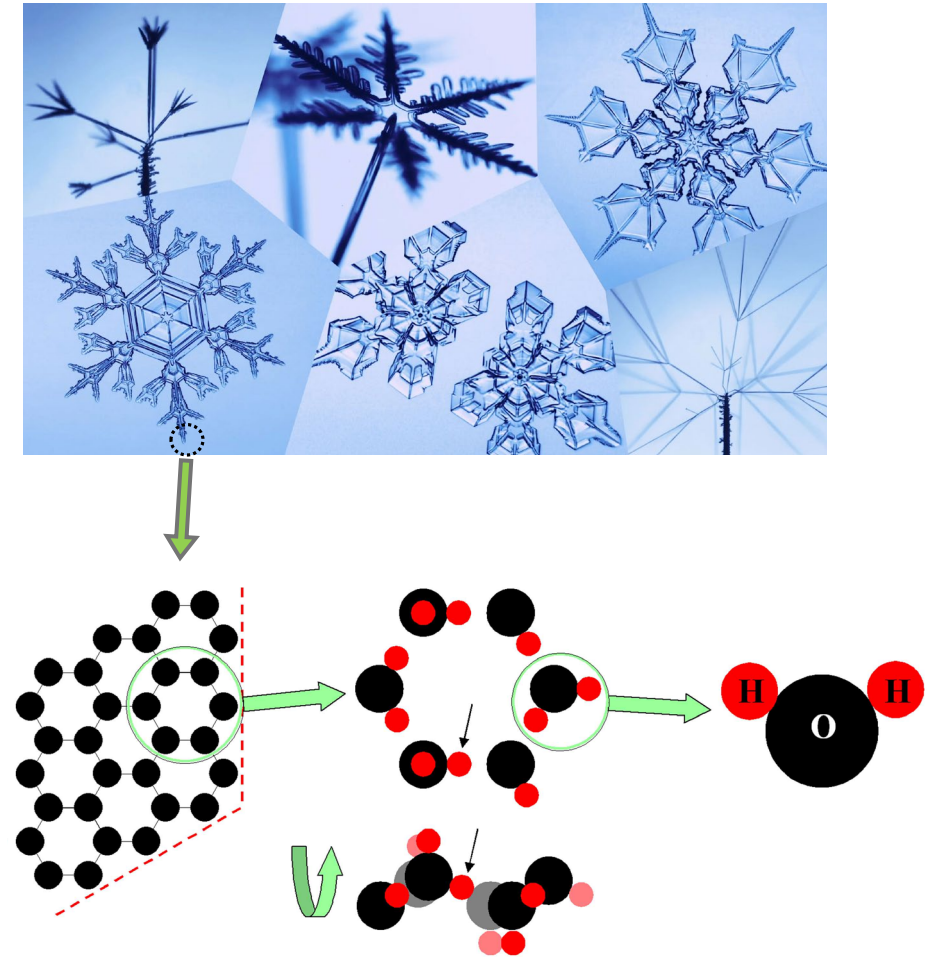


- Many properties of solid materials depends on their crystallinity and/or crystal structure



Crystal Structure

- ❑ **Crystal structure** is the **manner** in which atoms, ions, or molecules are **spatially arranged**
- ❑ Crystal structures of different materials can be relative simple or extremely complex
- ❑ To better describe crystal structure, atoms or ions are often thought of as solid spheres
- ❑ In **atomic hard-sphere model**, hard spheres representing **nearest**-neighboring atoms **touch** one another





Crystal Structures in Metals

- ▣ Metallic solids adopt metallic bonds
 - ▣ Non-directional
 - ▣ Electrons are highly delocalized
- minimal restriction of number and position of nearest-neighbor atoms
- ▣ Most metallic crystal structure adopt **dense (close) atomic packings**
 - ▣ **Face-centered cubic (FCC)** crystal structure
 - ▣ **Hexagonal close-packed (HCP)** crystal structure
- ▣ Some metals adopt non-close-packed structures
 - ▣ **Body-centered cubic (BCC)** crystal structure
 - ▣ **Simple cubic (SC)** structure



Packing and Close Packing in Metals

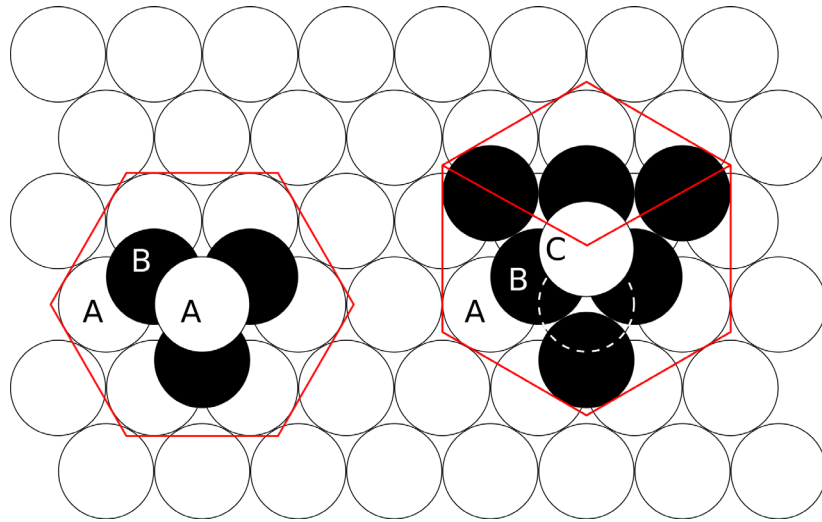


- In *two* dimensions, close packing refers to the **hexagonal** close-packed arrangement of spheres
- In *three* dimensions, close-packing structure can be achieved by stacking the two-dimensional close-packing layers
- **Lattice**: three-dimensional array of points coinciding with atom positions
- **Unit cell**: the smallest repeating unit of the structure which carries *all* the information necessary to construct unambiguously an infinite lattice
- **Coordination number** (for metals): number of **nearest**-neighbor atoms



Unit Cells of HCP and FCC Structures

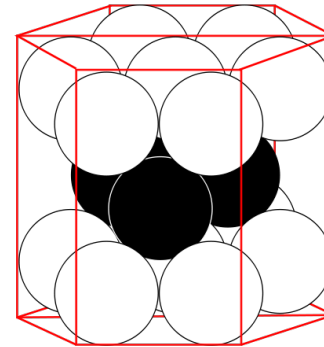
Top view:



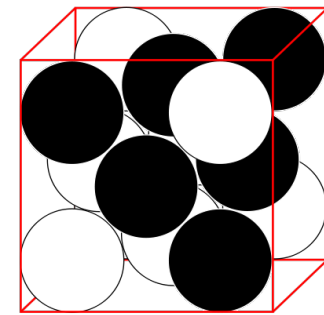
HCP

FCC (CCP)

3D view:



HCP



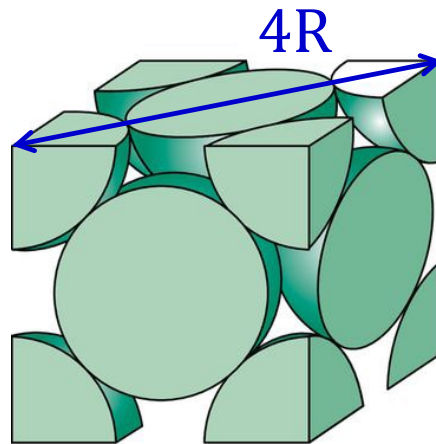
FCC (CCP)

- ▣ FCC structure is also called cubic-close-packed (CCP) structure

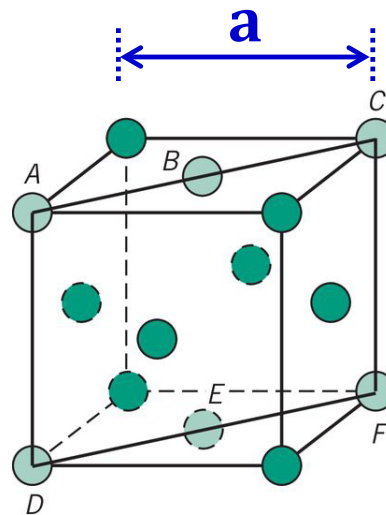


Unit Cell Representations of FCC Crystal Structure

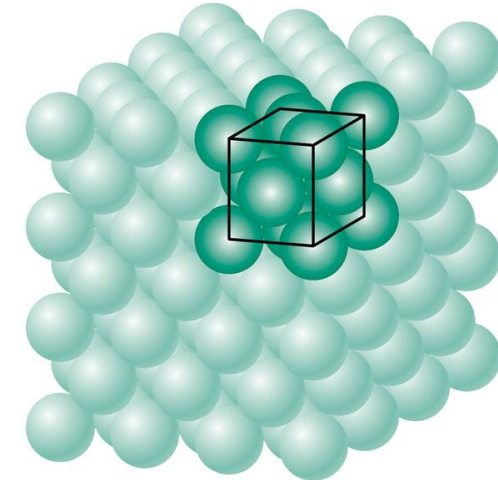
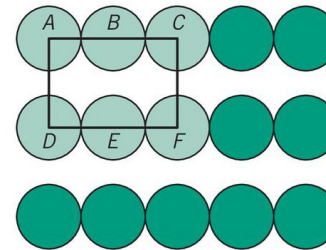
- ▣ Examples: Cu, Al, Ag, Au, Pt
- ▣ $a = 2\sqrt{2}R$
- ▣ Coordination number: 12



Hard-sphere
unit cell



Reduced-sphere
unit cell

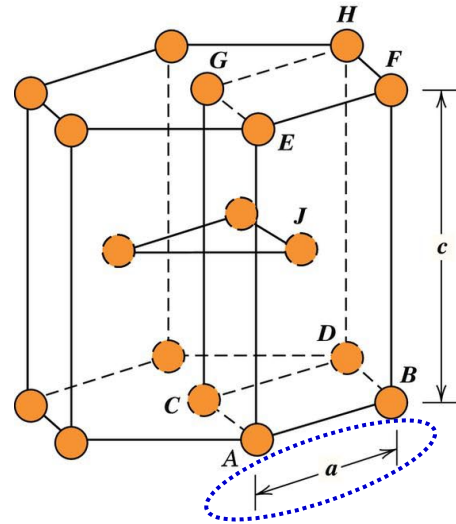


An aggregate of many atoms

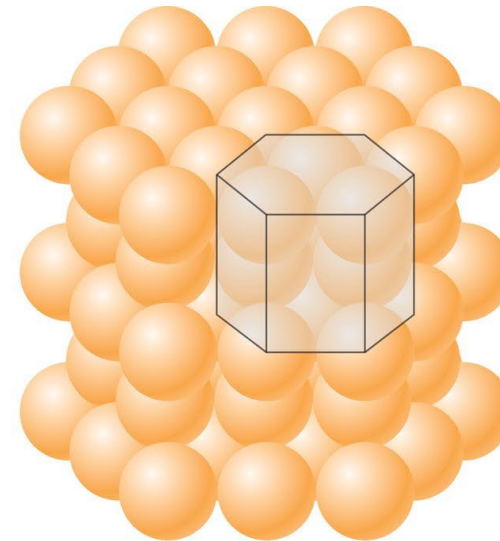


Unit Cell Representations of HCP Crystal Structure

- ▣ Examples: Co, Ti, Mg, Ru
- ▣ $a = 2R$
- ▣ Coordination number: 12



Reduced-sphere
unit cell

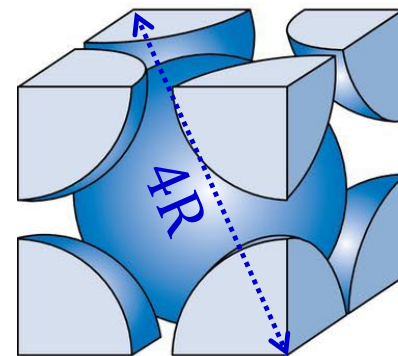
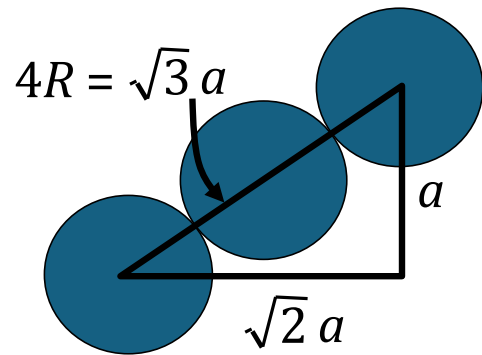


An aggregate of many atoms

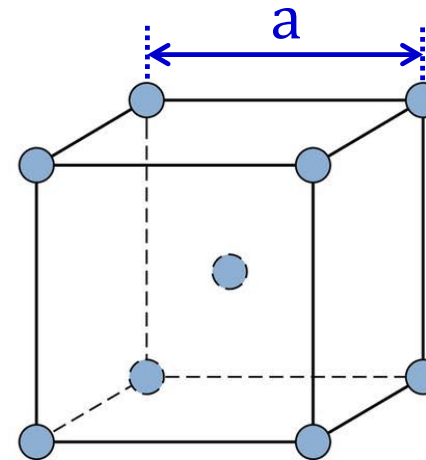


Unit Cell Representations of BCC Crystal Structure

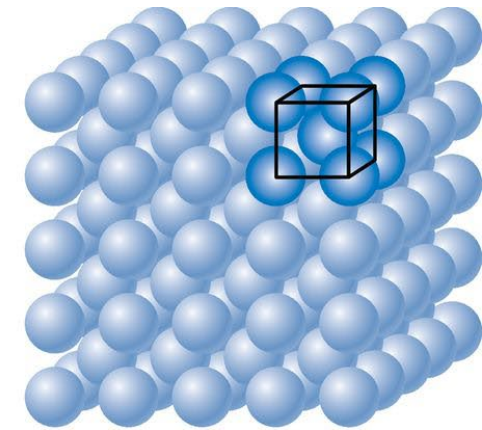
- ▣ Examples: Fe, Cr, W, Mo
- ▣ $a = \frac{4R}{\sqrt{3}}$
- ▣ Coordination number: 8



Hard-sphere
unit cell



Reduced-sphere
unit cell

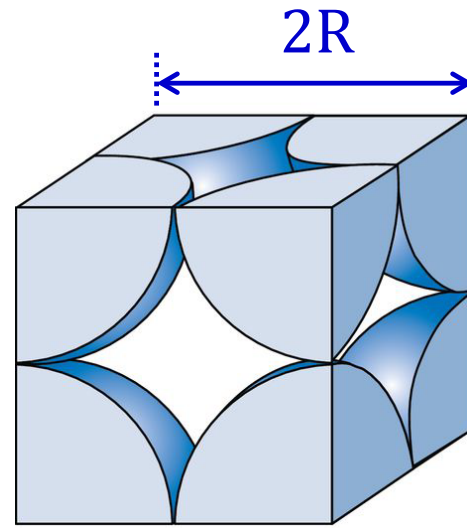


An aggregate of
many atoms

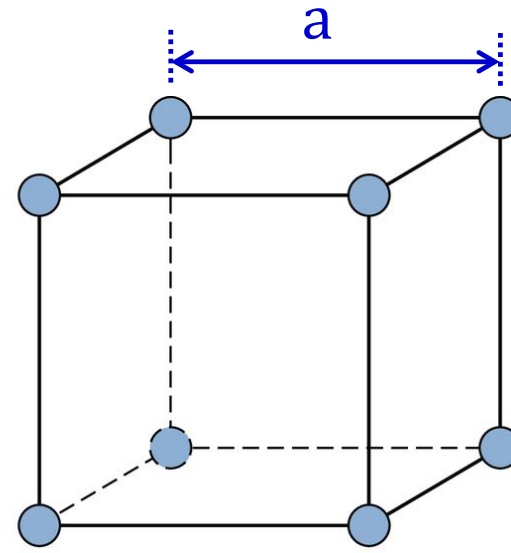


Unit Cell Representations of SC Crystal Structure

- ▣ Examples: Po
- ▣ $a = 2R$
- ▣ Coordination number: 6



Hard-sphere unit cell



Reduced-sphere unit cell



Atomic Packing Factor (APF)

- ▣ **Atomic Packing Factor (APF):** sum of the **sphere volumes** of all atoms within a unit cell, assuming the atomic hard-sphere model, **divided by** the **unit cell volume**

$$\text{APF} = \frac{\text{volume of atoms actually inside the unit cell}}{\text{total volume of unit cell}}$$

- ▣ To calculate APF, we need to first find out **number** of atoms inside one unit cell, then calculate the total volume
- ▣ Recall the volume of each sphere is $\frac{4\pi R^3}{3}$



Exercise



Show that in an FCC structure, the APF is approximately 0.74.



Exercise

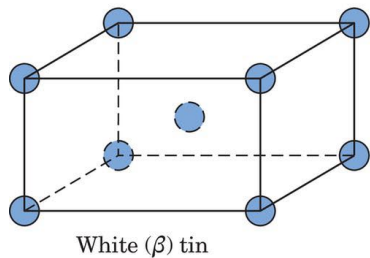


Copper has an atomic radius of 0.128 nm, an FCC crystal structure, an atomic weight of 63.5 g/mol, and a measured density of 8.94 g/cm³. Compute its theoretical density in g/cm³.



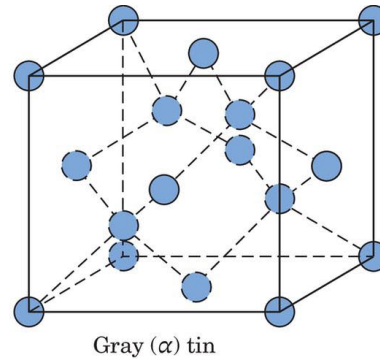
Polymorphism and Allotropy

- ▣ Some metals and non-metals have *more than one* crystal structure
- ▣ This is **polymorphism** or **allotropy**
- ▣ The prevailing crystal structure of a metal depends on *temperature* and *pressure*
- ▣ Example: Sn

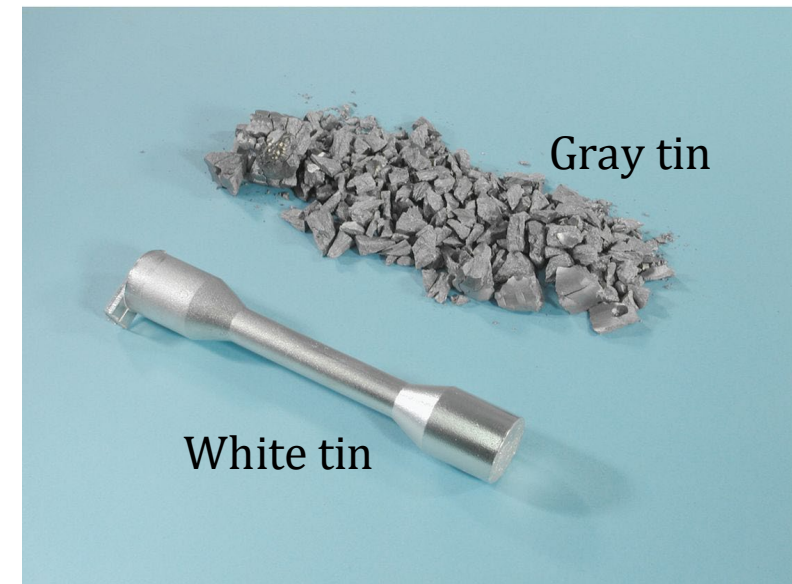


Body-centered tetragonal
crystal structure

13.2°C
Cooling



Diamond cubic
crystal structure

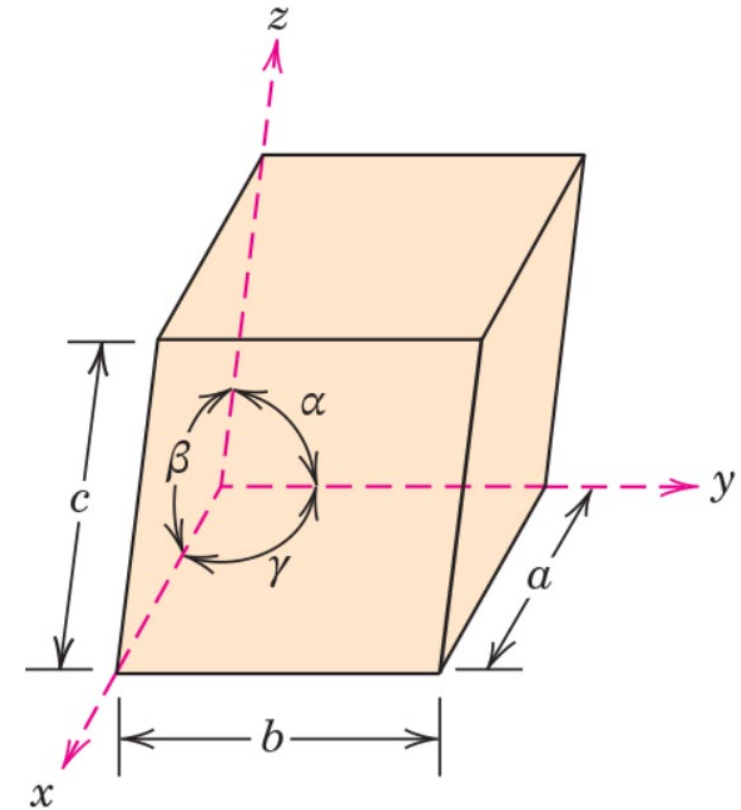


Photograph courtesy of Professor Bill Plumbridge, Department of Materials Engineering, The Open University, Milton Keynes, England.



Crystal Systems

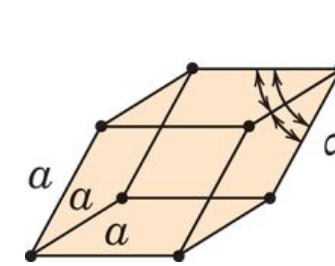
- ▣ There are many different possible crystal structures
- ▣ We can group all crystal structures based on **unit cell geometry**
- ▣ The geometry of a unit cell can be *completely* defined in terms of **6 parameters**:
 - ▣ 3 **edge lengths** a , b , and c
 - ▣ 3 **interaxial angles** α , β , and γ



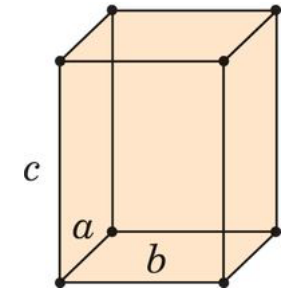


Seven Crystal Systems

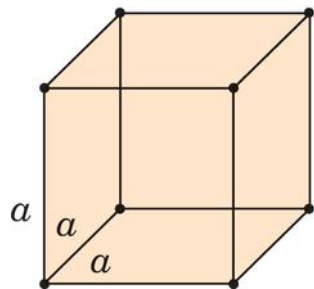
Crystal system	Axial relationships	Interaxial angles
Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$
Hexagonal	$a = b \neq c$	$\alpha = \beta = 90^\circ, \gamma = 120^\circ$
Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90^\circ$
Rhombohedral (trigonal)	$a = b = c$	$\alpha = \beta = \gamma \neq 90^\circ$
Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$
Monoclinic	$a \neq b \neq c$	$\alpha = \beta = 90^\circ \neq \gamma$
Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$



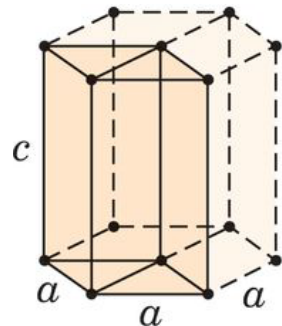
Rhombohedral



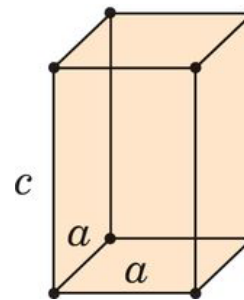
Orthorhombic



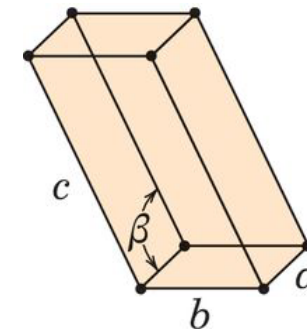
Cubic



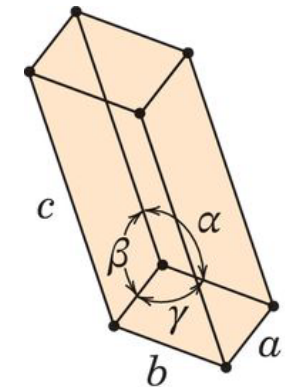
Hexagonal



Tetragonal



Monoclinic



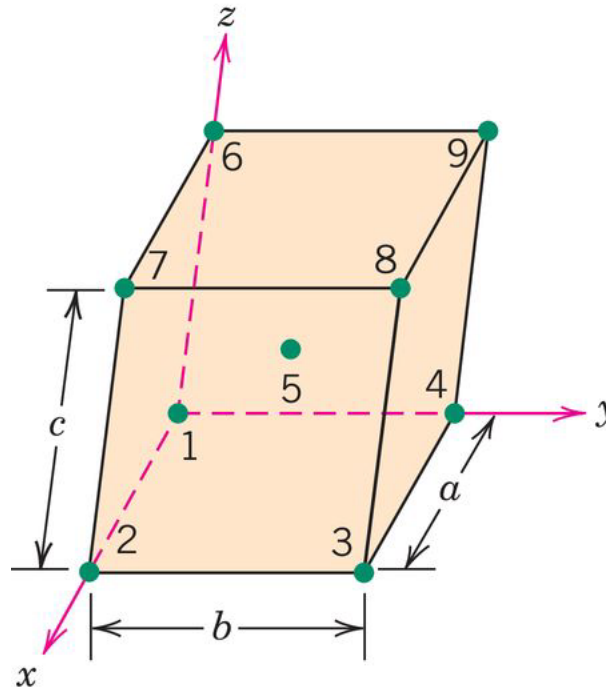
Triclinic



Crystallographic Points

- ▣ To specify a position within a unit cell, **lattice position coordinates** are used, specifically 3 **point indices**: q, r, s
- ▣ They are *fractional multiples* of unit cell lengths

- ▣ For example, in terms of crystallographic points, the positions of atoms in a body-centered unit cell can be expressed as:



Point number	q	r	s
1	0	0	0
2	1	0	0
3	1	1	0
4	0	1	0
5	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
6	0	0	1
7	1	0	1
8	1	1	1
9	0	1	1



Crystallographic Directions

- ▣ To describe a **crystallographic direction**, a *vector containing three dimensional indices* is used.
- ▣ Steps to determine the three directional indices:
 1. Place a unit cell at the corner of a right-handed x-y-z coordinate system
 2. Find the coordinates of two points lying on the direction:
 - e.g. Point 1 (x_1, y_1, z_1) as vector tail, Point 2 (x_2, y_2, z_2) as vector head
 3. Tail point coordinates are subtracted from head point coordinates
 - we get ($x_2 - x_1, y_2 - y_1, z_2 - z_1$)



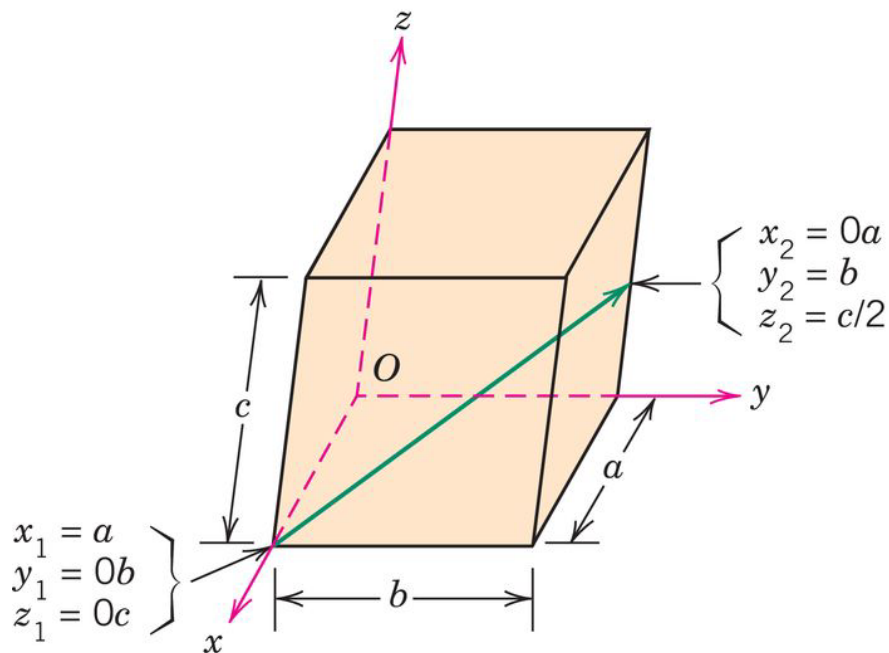
Crystallographic Directions

- ▣ Steps to determine the three directional indices (continued):
 4. The coordinate differences are **normalized** in terms of (*i.e.* divided by) their respective **a**, **b**, and **c** lattice parameters
 - we get $(\frac{x_2-x_1}{a}, \frac{y_2-y_1}{b}, \frac{z_2-z_1}{c})$
 5. For convenience, the three numbers are often multiplied/divided by a common factor (**n**) to change to **smallest integer** values
 6. The final indices, not separated by commas, are enclosed in **square brackets**:
[uvw]
 - $u = n(\frac{x_2-x_1}{a}), v = n(\frac{y_2-y_1}{b}), w = n(\frac{z_2-z_1}{c})$



Crystallographic Directions

Example: procedure to determine the following directional indices



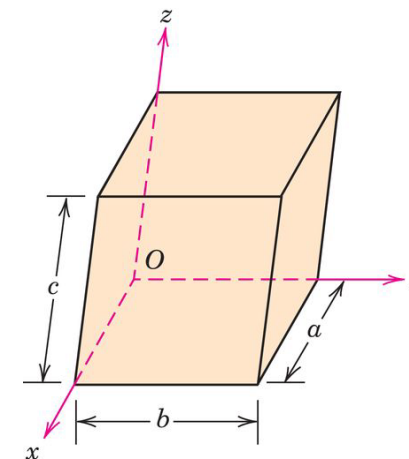
	x	y	z
Head coordinates (x ₂ , y ₂ , z ₂)	0	b	c/2
Tail coordinates (x ₁ , y ₁ , z ₁)	a	0	0
Coordinate differences	-a	b	c/2
Calculated values of u, v, w	u = -2	v = 2	w = 1
Final expression	[2̄21]		



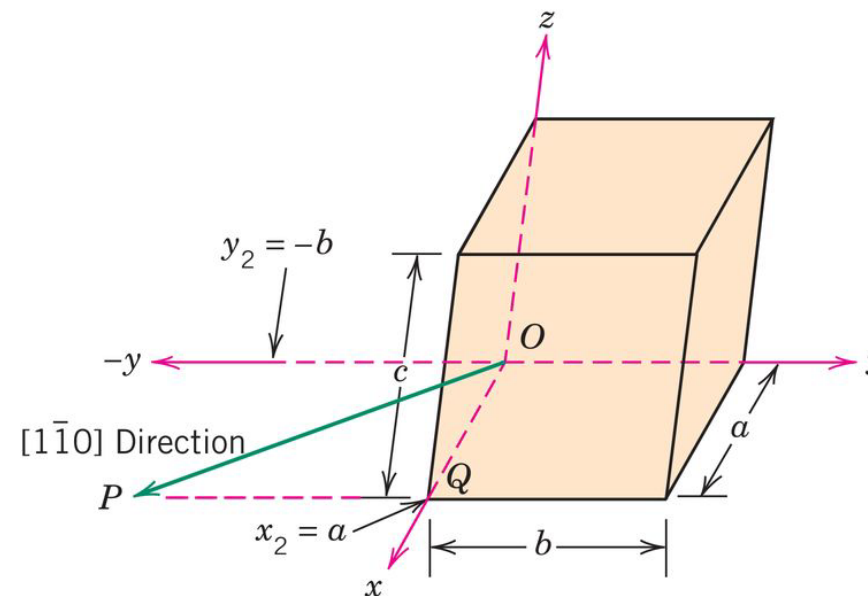
Crystallographic Directions

Example: construct a *specified* crystallographic direction

- Within the following unit cell, draw a $[1\bar{1}0]$ direction with its tail located at the origin of the coordinate system, Point O
- This problem can be solved by reversing the aforementioned procedure



	x	y	z
Head coordinates (x_2, y_2, z_2)	a	$-b$	0
Tail coordinates (x_1, y_1, z_1)	0	0	0
Coordinate differences	a	$-b$	0
Calculated values of u, v, w	$u = 1$	$v = -1$	$w = 0$
Final expression	$[1\bar{1}0]$		





Equivalent Crystallographic Directions

- ▣ For some crystal structures, several nonparallel directions with different indices are **crystallographically equivalent**
 - ▣ *e.g.*, in cubic crystals, all the directions represented by the following indices are equivalent: $[100]$, $[\bar{1}00]$, $[010]$, $[0\bar{1}0]$, $[001]$, $[00\bar{1}]$
- ▣ As a convenience, equivalent directions are grouped together into a **family**, which is enclosed in **angle brackets**, thus **$\langle uvw \rangle$** (*e.g.* $\langle 100 \rangle$)
- ▣ In cubic system, directions having the same indices *without* regard to order or sign, are equivalent
 - ▣ *e.g.* in cubic crystals, $[123]$ and $[\bar{2}1\bar{3}]$ are equivalent
- ▣ In other crystal systems, this is not always true
 - ▣ *e.g.* for tetragonal symmetry, $[100]$ and $[010]$ are equivalent, but $[100]$ and $[001]$ are *not* equivalent



Crystallographic Planes

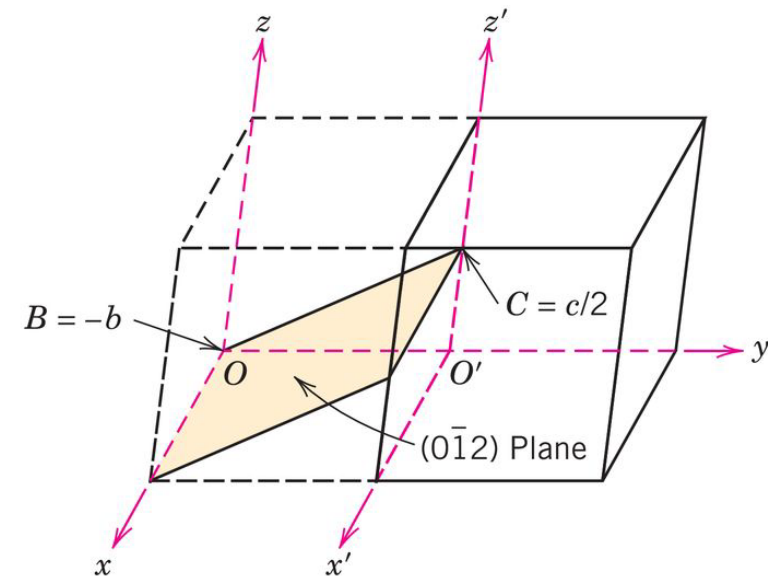
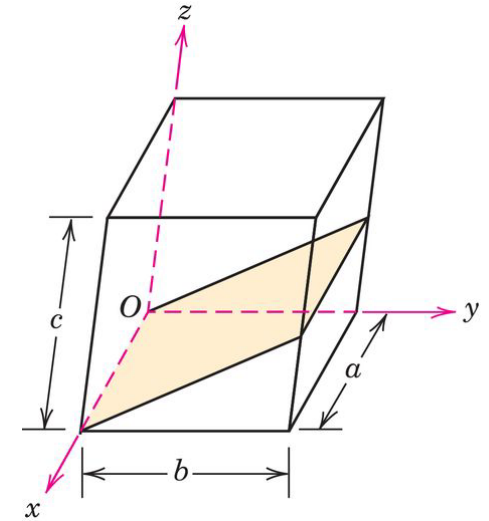
- ▣ Similar to directions, crystallographic planes are also represented by three indices, known as **Miller indices**
- ▣ Steps to determine the three Miller indices:
 1. If the plane passes through the origin, establish a new origin in another unit cell
 2. Read values of **intercepts** of plane (A, B, C) with x, y , and z axes **in terms of a, b, c**
 3. Take **reciprocals** of intercepts ($1/A, 1/B, 1/C$)
 4. **Normalize** them by multiplying by lattice parameters a, b , and c ($a/A, b/B, c/C$)
 5. Reduce to **smallest integer** values ($na/A, nb/B, nc/C$)
 6. Enclose resulting **Miller Indices** in **round brackets**, no commas: **(hkl)**



Crystallographic Planes

Example: determine the Miller indices for the plane shown on the right

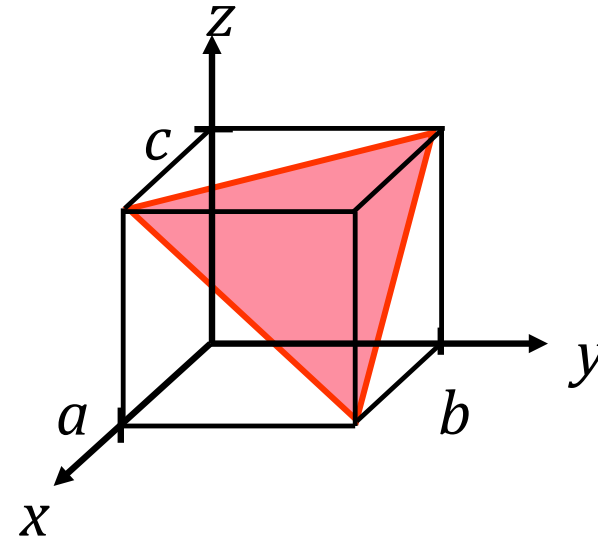
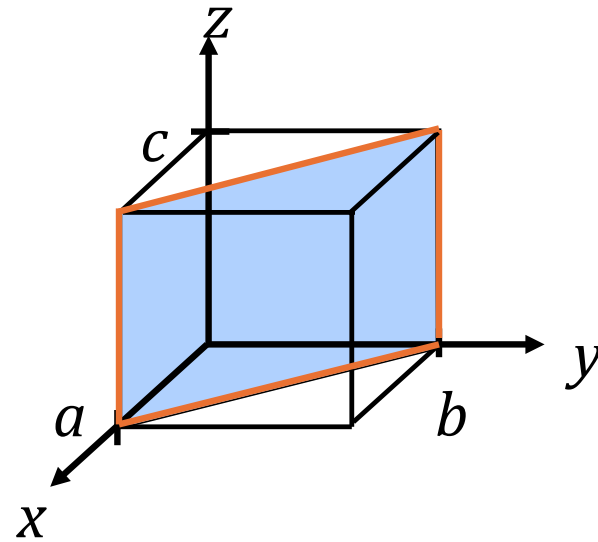
1. The plane passes through origin O , we choose a new origin O' in adjacent unit cell
2. Intercepts (A, B, C) of plane:
3. Take reciprocals of intercepts:
4. Normalize them by multiplying by lattice parameters a, b , and c :
5. Reduce to smallest integer values:
6. Enclosure of Miller Indices:





Exercise

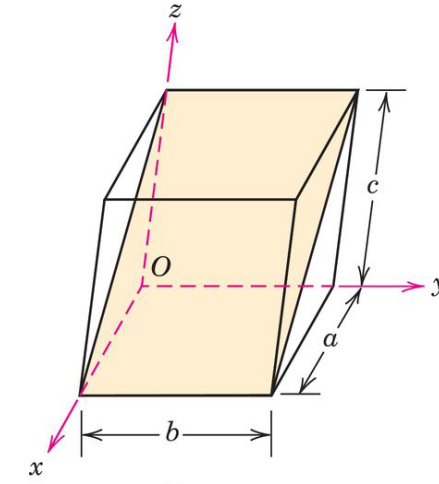
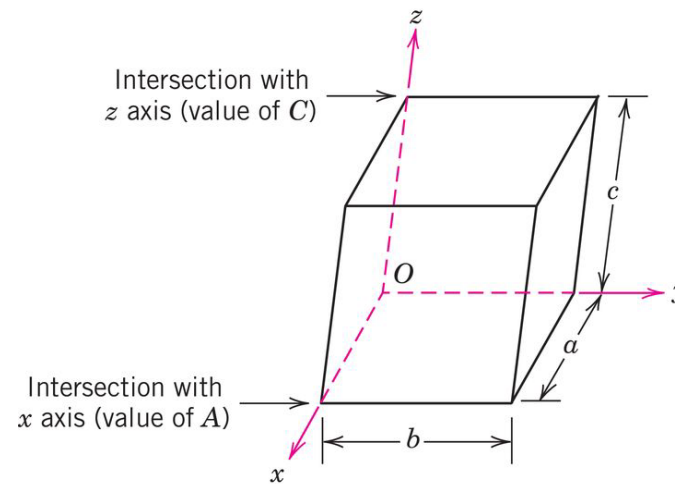
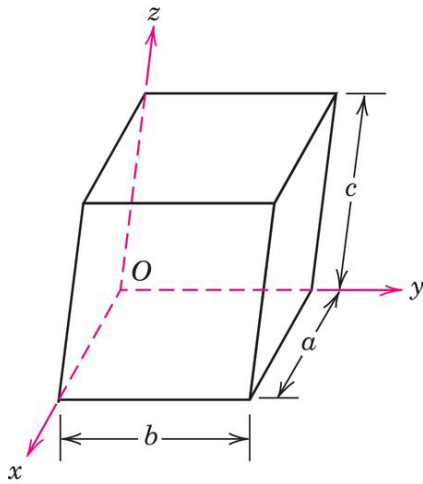
Quick test: what are the Miller indices for the planes shown below:





Crystallographic Planes

Example: construct a (101) plane within the following unit cell



$$h = \frac{na}{A} = 1$$

$$k = \frac{nb}{B} = 0$$

$$l = \frac{nc}{C} = 1$$

already are the smallest integer so $n = 1$

$$A = a$$

$$B = \infty$$

$$C = c$$

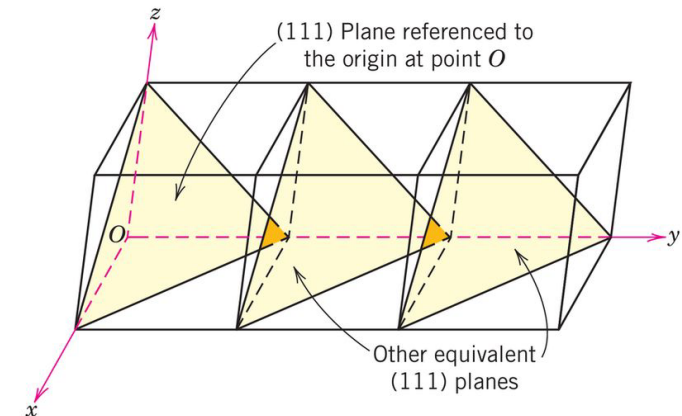
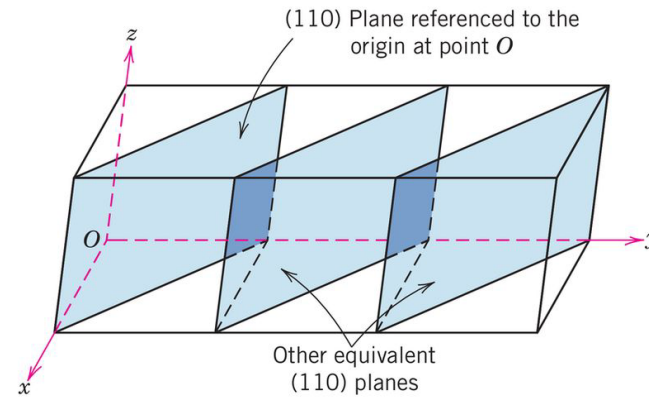
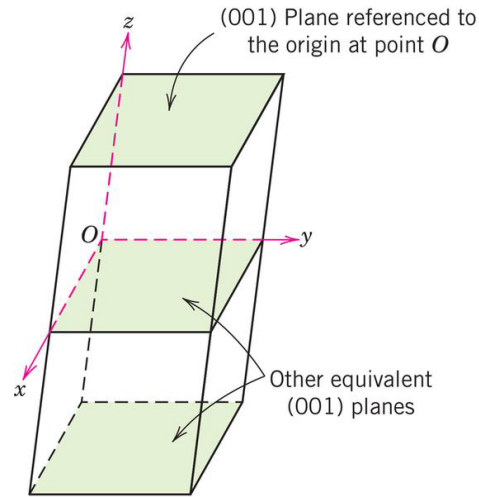
Therefore, (101) plane

- Intersects the x axis at a
- Parallel to y axis
- Intersects the z axis at c



Equivalent Crystallographic Planes

- Any two planes parallel to each other are **equivalent** and have **identical indices**

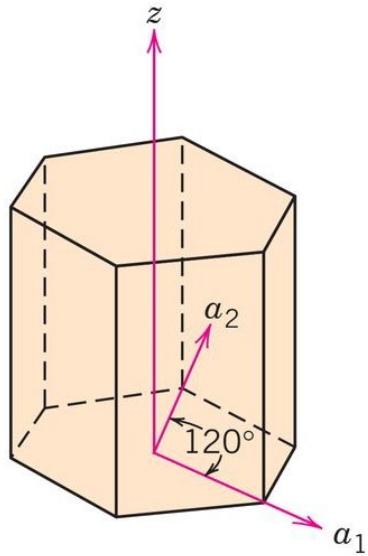


- A family of crystallographically equivalent planes is denoted in **curly brackets**, **$\{hkl\}$**
 - e.g.*, in cubic crystals, (111) , $(\bar{1}11)$, $(1\bar{1}1)$, $(11\bar{1})$, $(1\bar{1}\bar{1})$, $(\bar{1}\bar{1}1)$, $(\bar{1}\bar{1}\bar{1})$, and $(\bar{1}\bar{1}\bar{1})$ planes all belong to the $\{111\}$ family



Directions and Planes in Hexagonal Crystals

- There are two ways to express directions and planes in hexagonal unit cell
- Three-index** system: $[UVW]$
- Four-index** system (Miller-Bravais coordinate system): $[uvw]$



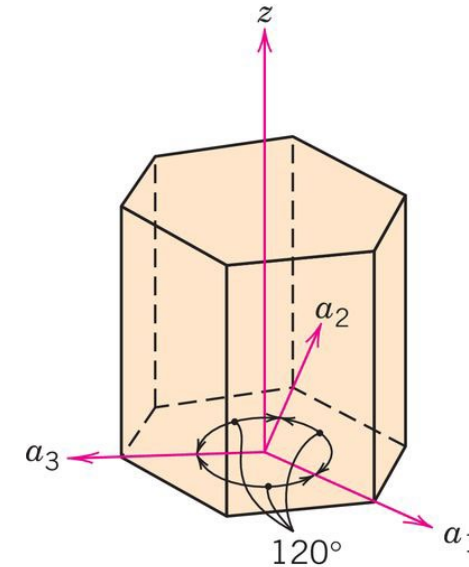
Conversion formula:

$$u = \frac{1}{3} (2U - V)$$

$$v = \frac{1}{3} (2V - U)$$

$$t = -(u + v)$$

$$w = W$$



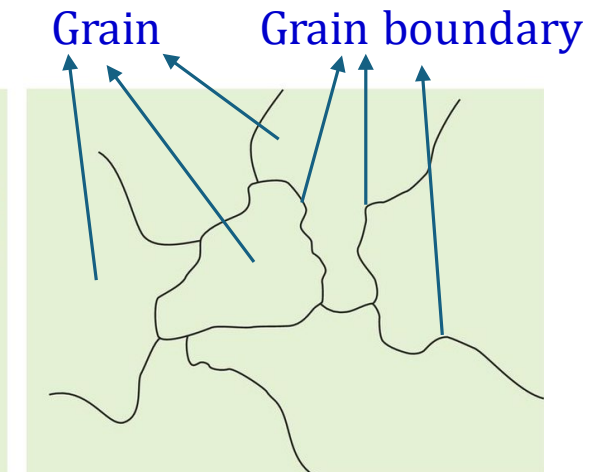
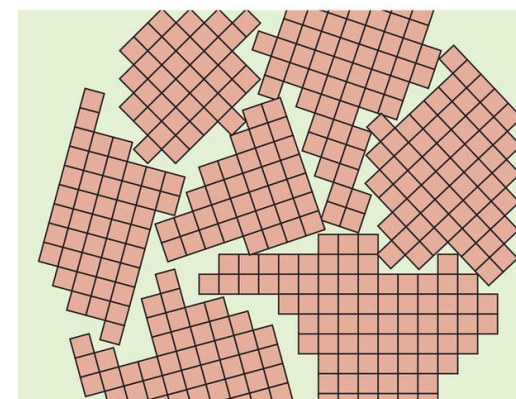
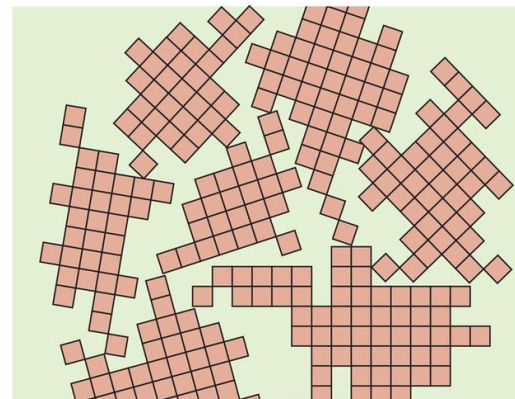
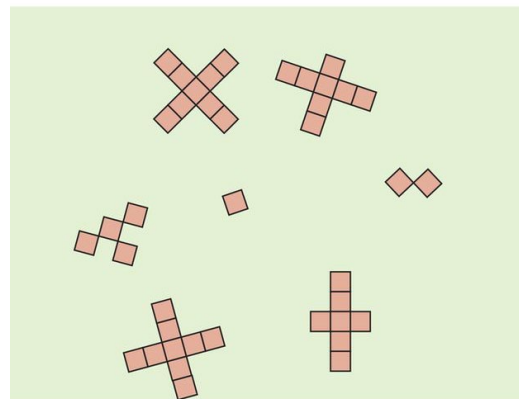


Single Crystal and Polycrystalline Materials

- ▣ If the unit cell can repeat itself in a crystalline solid without interruption, the result is a **single crystal**
- ▣ A single crystal usually assumes a regular geometric shape with flat faces; the shape is indicative of the crystal structure
- ▣ Most crystalline solids are composed of many small single crystals, or **grains**; they are **polycrystalline**



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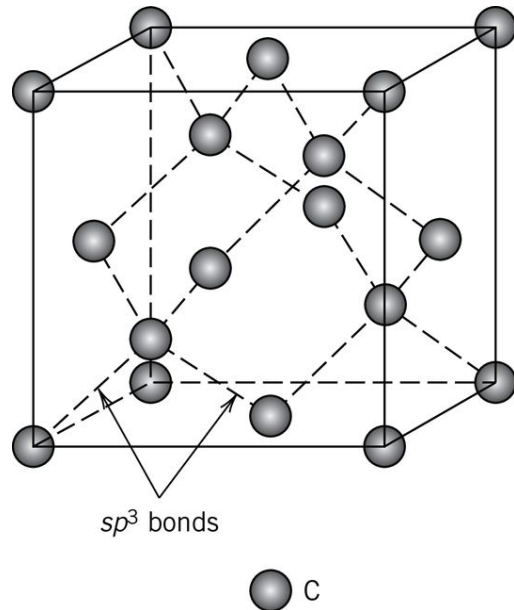
Stages of solidification of a polycrystalline material



Some Important Structure Types

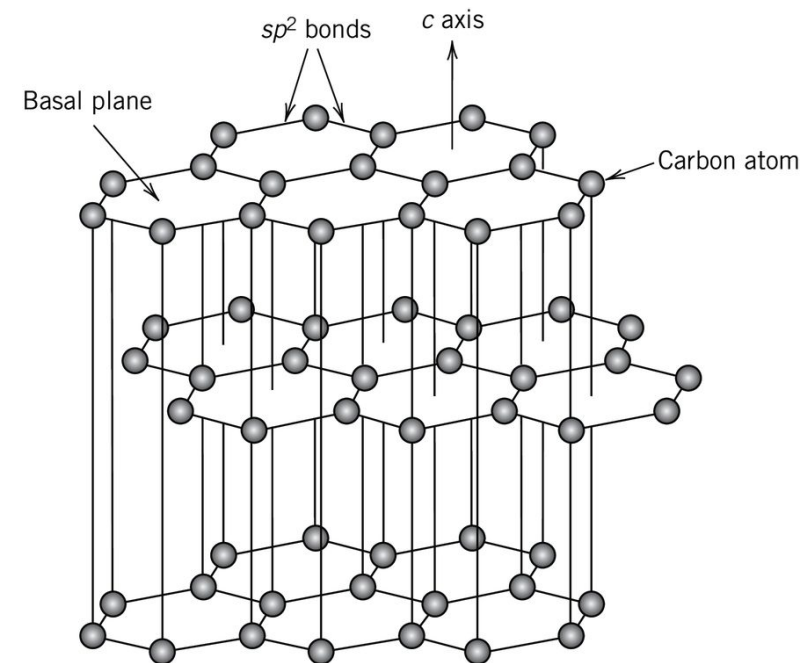
▣ Carbon

▣ *Diamond*



e.g. C, Si, Ge

▣ *Graphite*

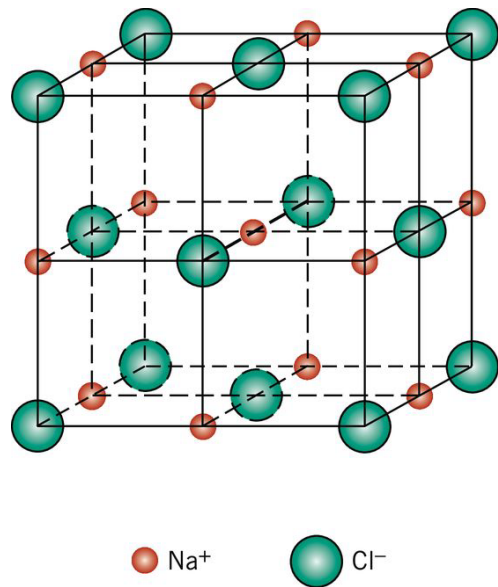




Some Important Structure Types

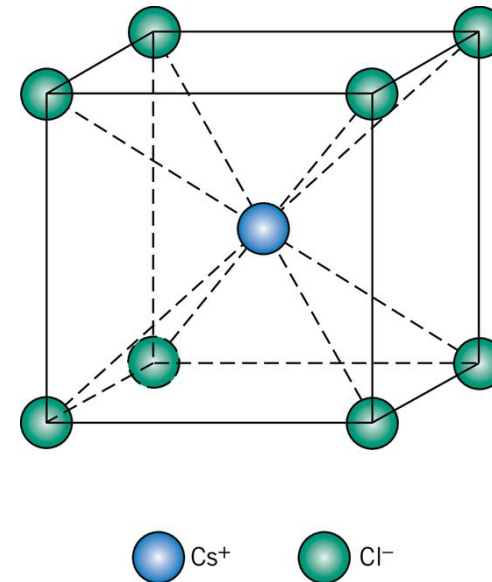
▣ AX-Type Crystal Structures

▣ *Rock Salt Structure*



e.g. NaCl, MgO, FeO

▣ *Cesium Chloride Structure*



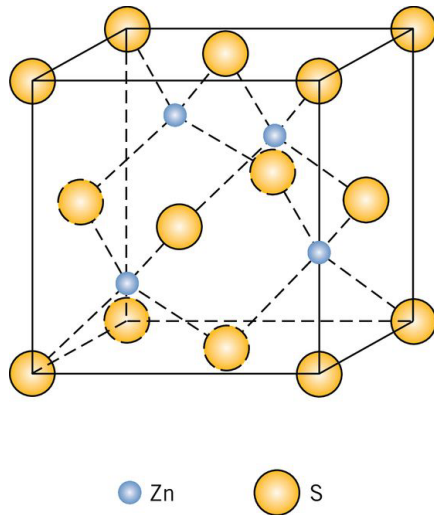
e.g. CsCl, CsBr



Some Important Structure Types

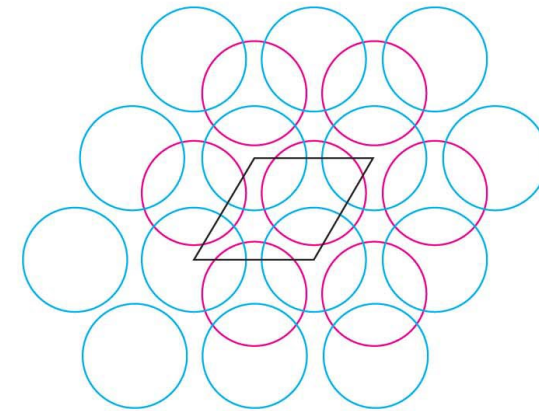
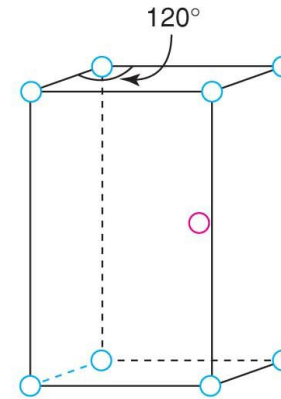
▣ AX-Type Crystal Structures

▣ *Zinc Blende (sphalerite)*



e.g. ZnS, SiC, GaAs

▣ *Wurtzite*



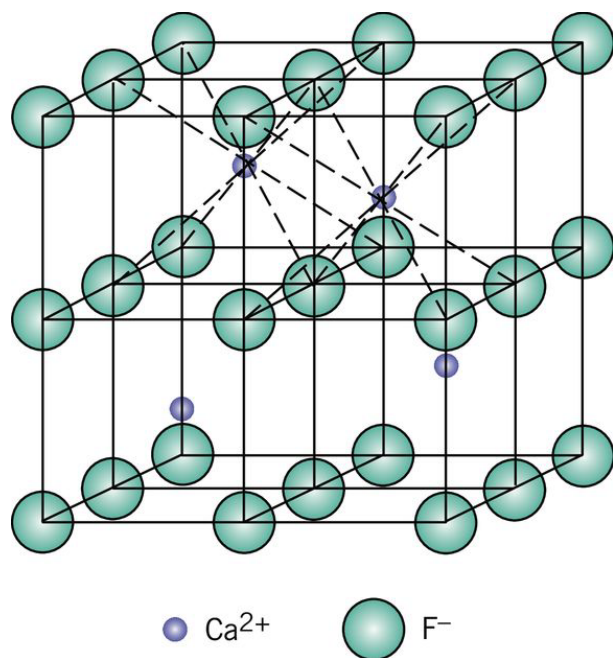
e.g. ZnS, ZnO, CdS, CdSe



Some Important Structure Types

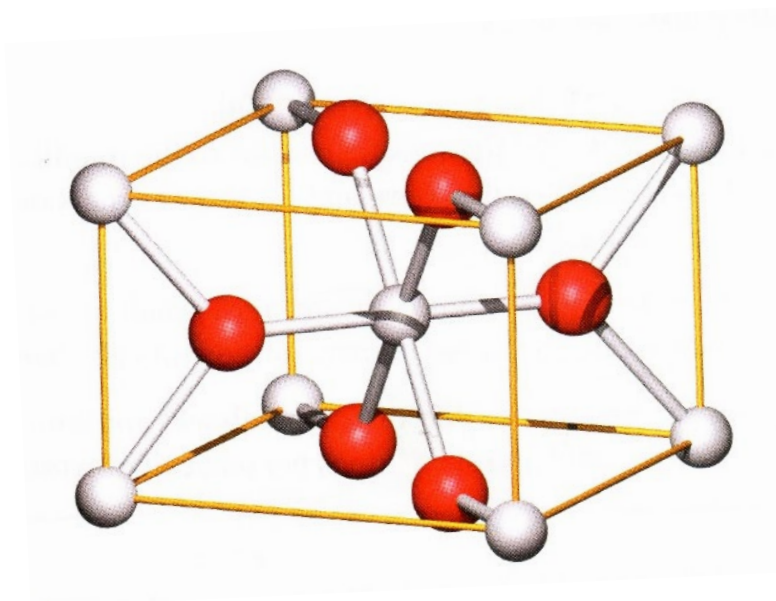
▣ AX_2 -Types Crystal Structures

▣ *Fluorite*



e.g. CaF_2 , UO_2 , ThO_2 ,

▣ *Rutile*

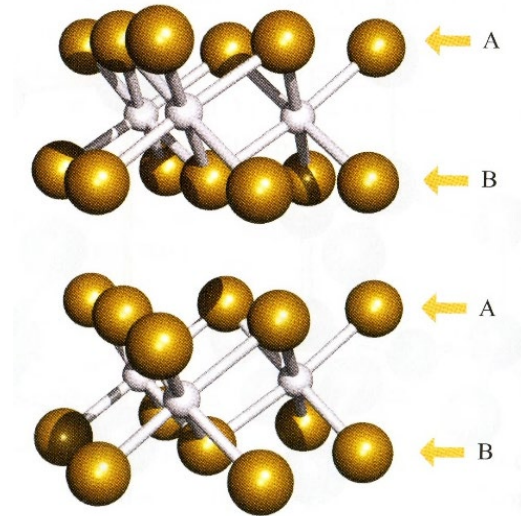


e.g. TiO_2 , RuO_2 , IrO_2



Some Important Structure Types

- ▣ **AX_2 -Types Crystal Structures**
 - ▣ ***CdI₂ Structure*** (one kind of layer structures)



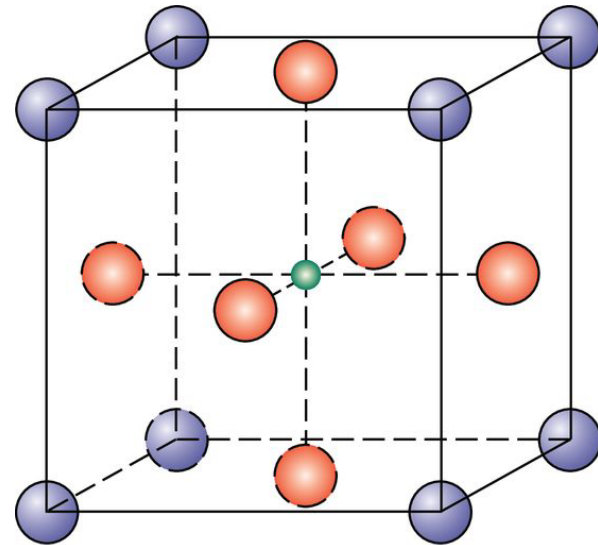
e.g. CdI₂, FeBr₂, Mg(OH)₂, Co(OH)₂



Some Important Structure Types

▣ ABX_3 -Type Crystal Structures

▣ *Perovskite*



e.g. $BaTiO_3$, $SrZrO_3$, $SrSnO_3$